Chapter I. Random Signals

I.1 Random variables

Random variables

- A **random variable** is a variable that holds a value produced by a (partially) random phenomenon
  - basically it is *a name* attached to an arbitrary value
  - short notation: r.v.
- Typically denoted as X, Y etc..
- Examples:
  - X = The value of a dice
  - V_{in} = The value of the voltage in one point of a circuit

Glossary

- “*i.e.*” = *id est* = “that is” = “adică”
- “*e.g.*” = *exempli gratia* = “for example” = “de exemplu”

Realizations

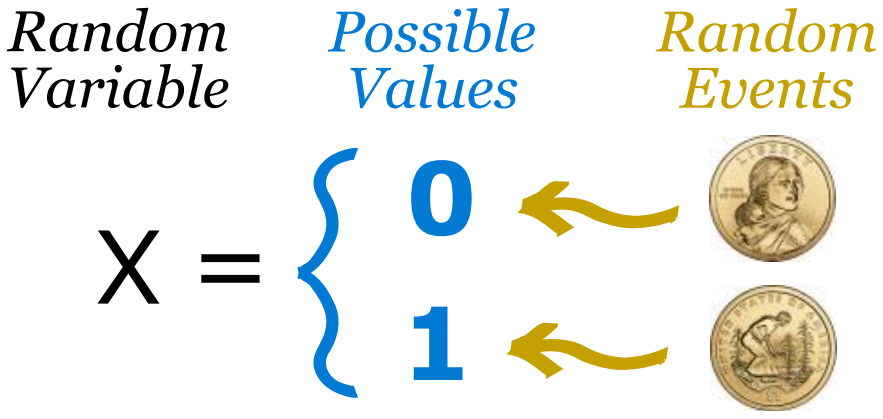
- **A realization** of a random variable = one possible value it can take
  - e.g. the value 3 of a dice
  - at different times, one may get different realizations
- Sample space Ω = the set of all values that can be taken by a random variable X
  - i.e. the set of all possible realizations
- Example: rolling a dice
  - The r.v. is denoted as X
  - We might get a realization $X = 6$
  - But we could have got any value from the sample space

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

- **Discrete** random variable: takes values from a discrete set
  - Example: value of a dice
- **Continuous** random variable: takes values from a continuous domain
  - Example: a voltage value

Throwing a coin

- Random variable X = "the face obtained by throwing a coin"



(image from *https://www.mathsisfun.com/data/random-variables.html*)

Why random variables?

- Random variables are a great model for **noise**
- Examples:
  - Measure a voltage in a circuit
  - Measure several times, the value is never precisely the same.

The value always *varies* a little. - i.e. it is affected by noise

Probability Mass Function

- Consider a **discrete** r.v. A
- The **probability mass function (PMF)** = the probability that A has value x

$$w_A(x) = P\{A = x\}$$

- Also known as the **distribution** of A
- Example: what is the PMF of a dice? Plot on board.
- Probability that A is equal to some value v

$$P\{A = v\} = w_A(v)$$

- Probability that A is between a and b (including):

$$P\{a \leq A \leq b\} = \sum_{x=a}^{b} w_A(x)$$

Cumulative Distribution Function

- The **cumulative distribution function (CDF)** = the probability that the value of A is smaller or equal than x

$$F_A(x) = P\{A \leq x\}$$

- In Romanian: *"funcție de repartitie"*
- Example: what is the CDF of a dice? Plot on board.
- For discrete r.v., the CDF is "stairwise"
- Probability that A is equal to some value v

$$P\{A = v\} = F_A(v) - F_A(v-1)$$

- Probability that A is between a and b (including):

$$P\{a \leq A \leq b\} = F_A(b) - F_A(a-1)$$

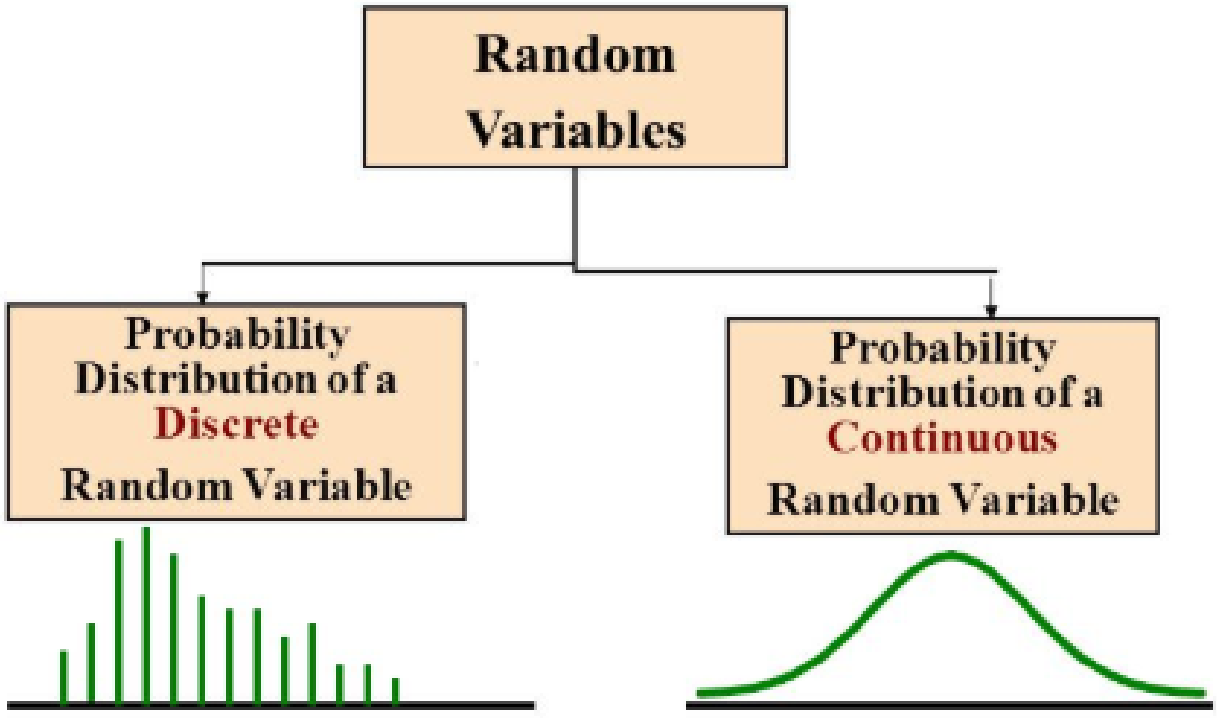
Relation between PMF and CDF

- CDF is the *cumulative sum* (i.e. the integral) of PMF

$$F_A(x) = \sum_{t=-\infty}^{t=x} w_A(t)$$

- Example for dice: easy to notice graphically

Continuous and discrete random variables



(image from "Probability Distributions: Discrete and Continuous", Seema Singh, *https://towardsdatascience.com/probability-distributions-discrete-and-continuous-7a94ede66dc0*)

Cumulative Distribution Function

- Consider a **continuous** r.v. A
- The **cumulative distribution function (CDF)** = the probability that the value of A is smaller or equal than x

$$F_A(x) = P\{A \leq x\}$$

- Same definition as for discrete random variables
- The **Probability Density Function (PDF)** of A (the **distribution** of A) is the derivative of the CDF:

$$w_A(x) = \frac{\mathrm{d}F_A(x)}{\mathrm{d}x}$$

- Informally, $w_A(x)$ is proportional to probability that the value of A is **around** x

Probability of an exact value

- The probability that a continuous r.v. A is **exactly** equal to a value x is **zero**
  - because there are an infinity of possibilities (continuous)
  - That's why we can't define a probability mass function like for discrete r.v.
- That's why the PDF says **in a small vicinity** around some value x, and not precisely equal to x

Computing probability based on PDF and CDF

- Probability that A is equal to some value v is always 0

$$P\{A = v\} = 0$$

- Probability that A is between a and b = integral of PDF from a to b:

$$P\{a \leq A \leq b\} = \int_a^b w_A(x)dx$$

- Probability that A is between a and b:

$$P\{a \leq A \leq b\} = F_A(b) - F_A(a)$$

- Doesn't matter if we consider closed or open interval
  - $[a, b]$ or (a, b)
  - why?

Relation between PDF and CDF

- CDF is **the integral** of PDF
- PDF is **the derivative** of CDF

$$F_A(x) = \int_{-\infty}^{x} w_A(u)\mathrm{d}u$$

$$\begin{aligned} w_A(x) &= \frac{\mathrm{d}F_A(x)}{\mathrm{d}x} \\ &= \lim_{\epsilon \to 0} \frac{F_A(x+\epsilon) - F_A(x-\epsilon)}{2\epsilon} \\ &= \lim_{\epsilon \to 0} \frac{P(A \in [x-\epsilon, x+\epsilon])}{2\epsilon} \end{aligned}$$

Graphical interpretation

- Probability that a continuous r.v. A is between a and b is **the area below the PDF**
  - i.e. the integral from a to b
- Probability that A is exactly equal to a certain value is zero
  - the area below a single point is zero

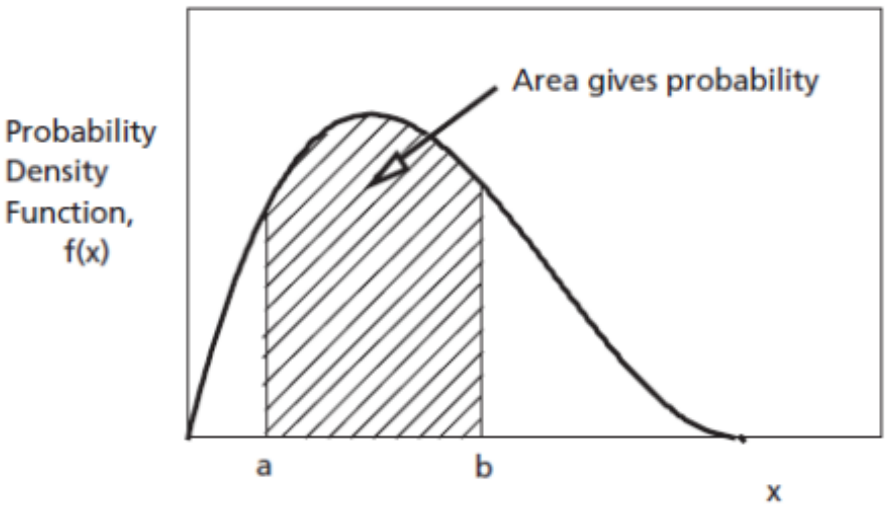


image from here

Discrete vs continuous r.v.

Comparison of discrete vs continuous random variables:

- The CDF $F_A(x)$ is defined identically, means same thing
- The PDF/PMF $w_A(x)$ is the derivative of CDF
  - for continuous r.v.:
    - it is a proper derivative
    - it means probability to be "around" x
  - for discrete r.v:
    - sort of "discrete derivative"
    - it means probability to be exactly equal to x

Properties of random variables

CDF:

- The CDF is always ≥ 0
- The CDF is always monotonously increasing (non-decreasing)
- The CDF starts from 0 and goes up to 1

$$F_A(-\infty) = 0 \quad F_A(\infty) = 1$$

PDF/PMF:

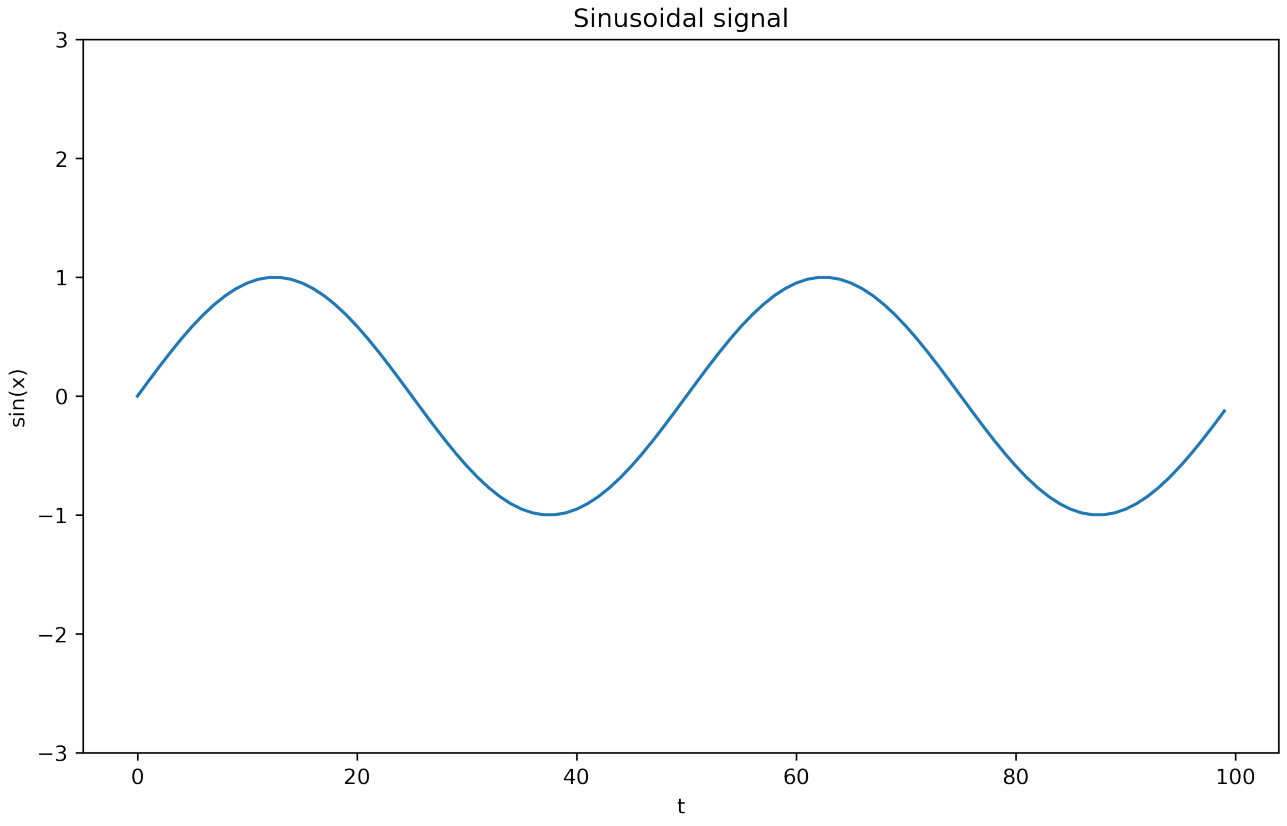
- The PDF/PMF are always ≥ 0
- Integral/sum over all of the PDF/PMF $= 1$

$$\int_{-\infty}^{\infty} w_A(x)\mathrm{d}x = 1$$

$$\sum_{x=-\infty}^{\infty} w_A(x) = 1$$

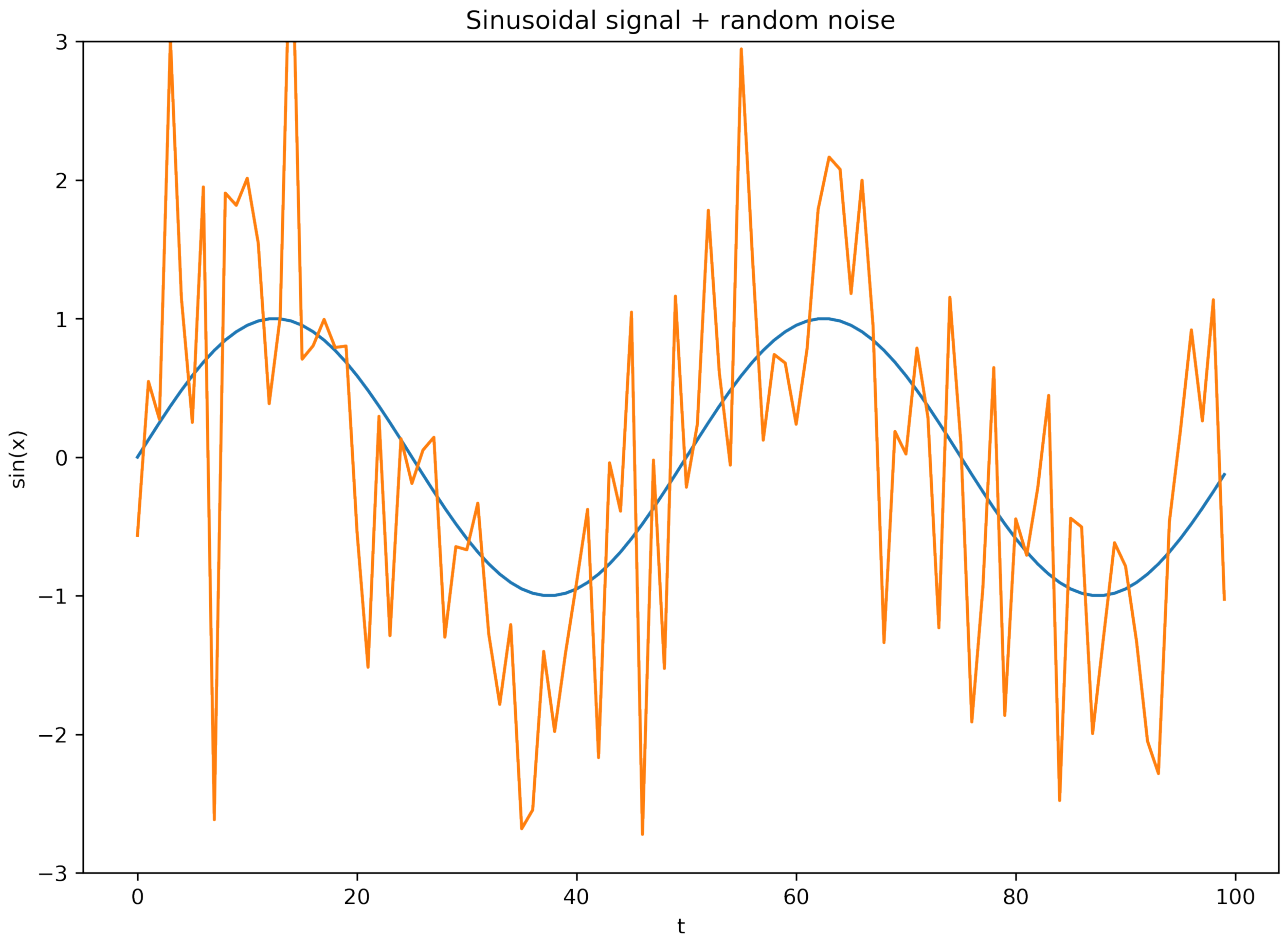
Various noise distributions

- Normal sine signal



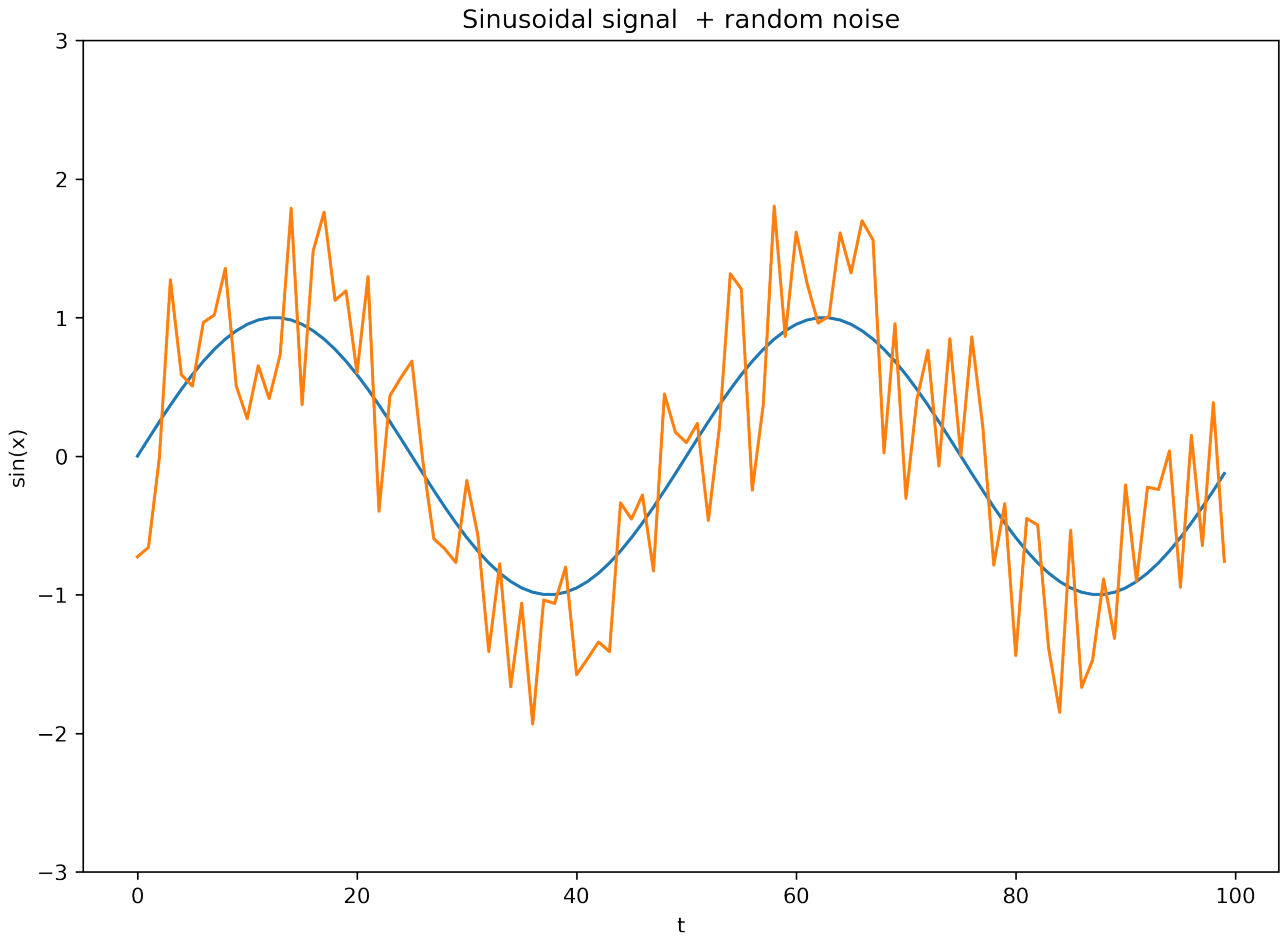
Various noise distributions

- Sine + noise 1 (normal, $\mu = 0, \sigma^2 = 1$)



Various noise distributions

- Sine + noise 2 (uniform $\mathcal{U}[-1, 1]$)
- What's different? The distribution type



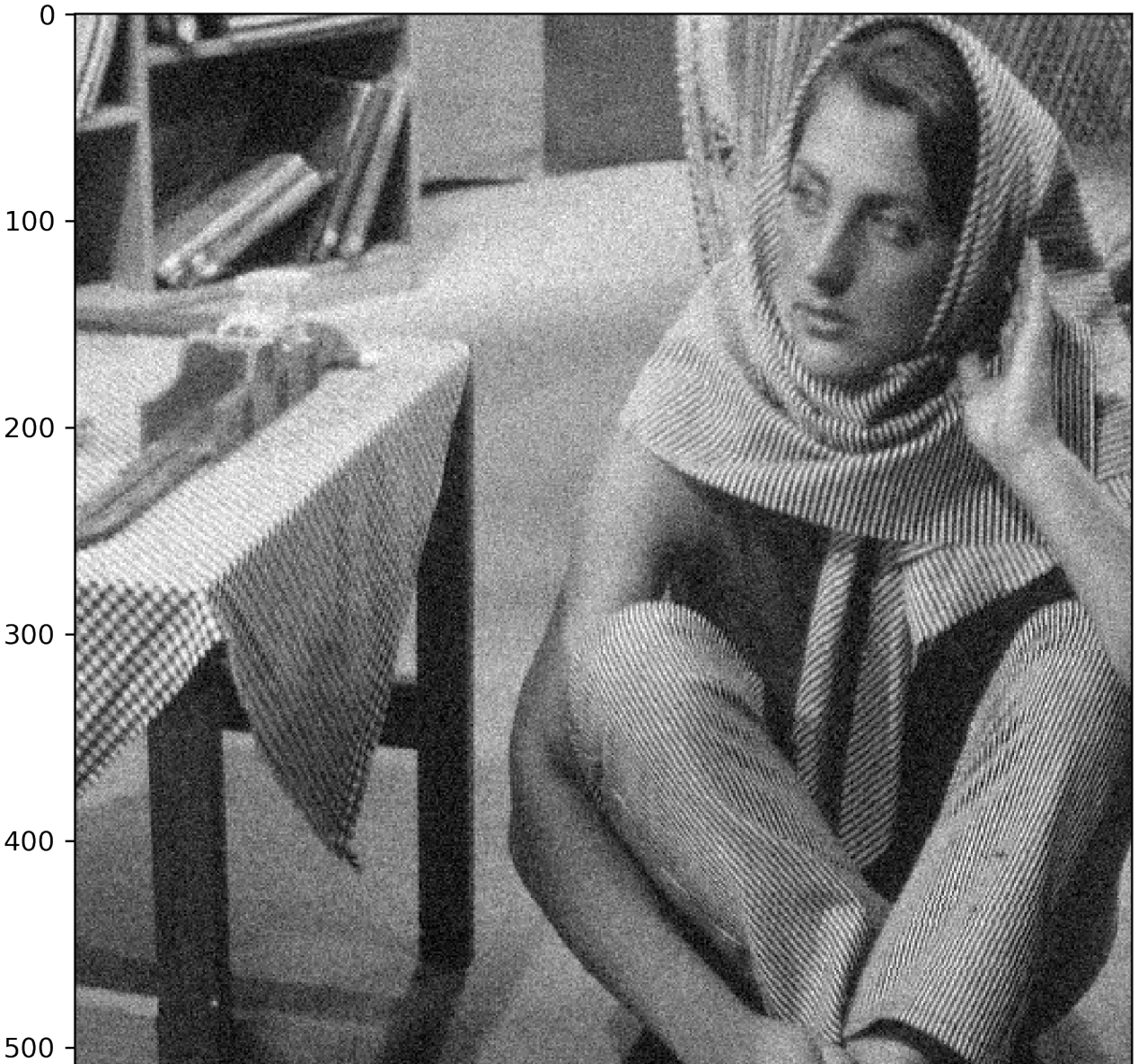
Various noise distributions

- Clean Image



Various noise distributions

- Image + noise (normal, $\mu = 0, \sigma^2 = 1$)



Various noise distributions

- Image + larger noise (normal, $\mu = 0, \sigma^2 = 10$)



Various noise distributions

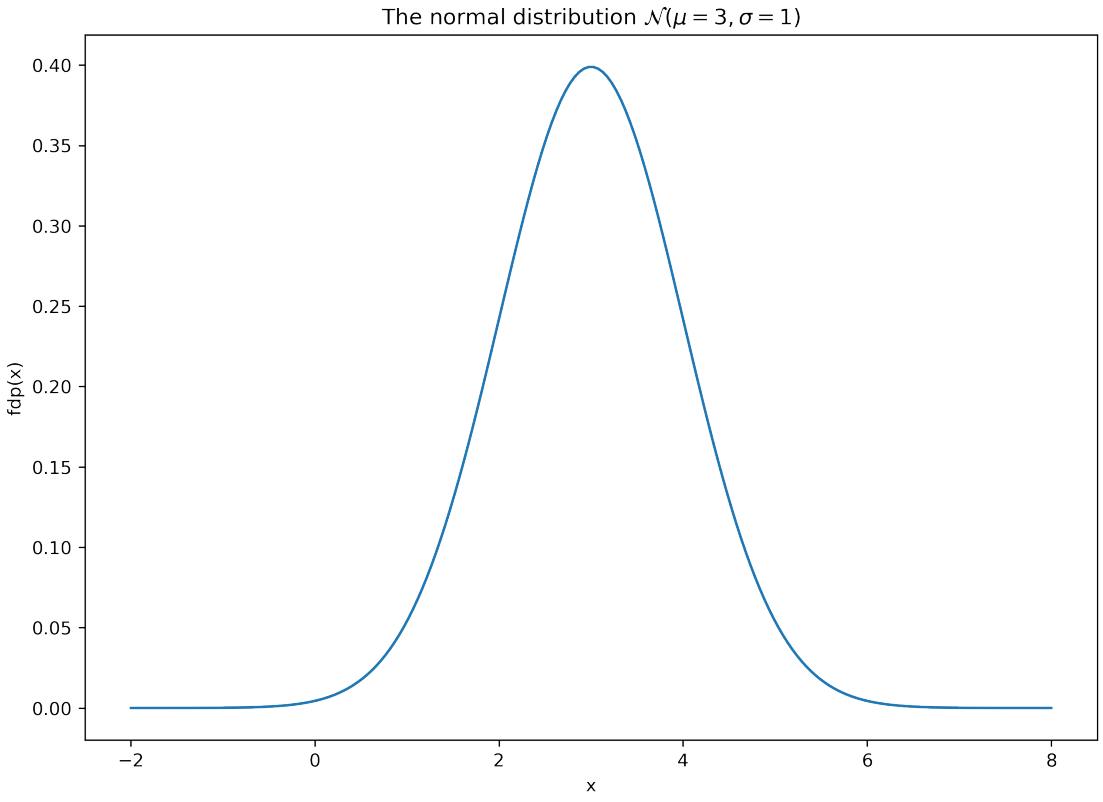
- Image + noise (uniform, $\mathcal{U}[-5, 5]$)



The normal distribution

- Probability density function

$$w_A(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



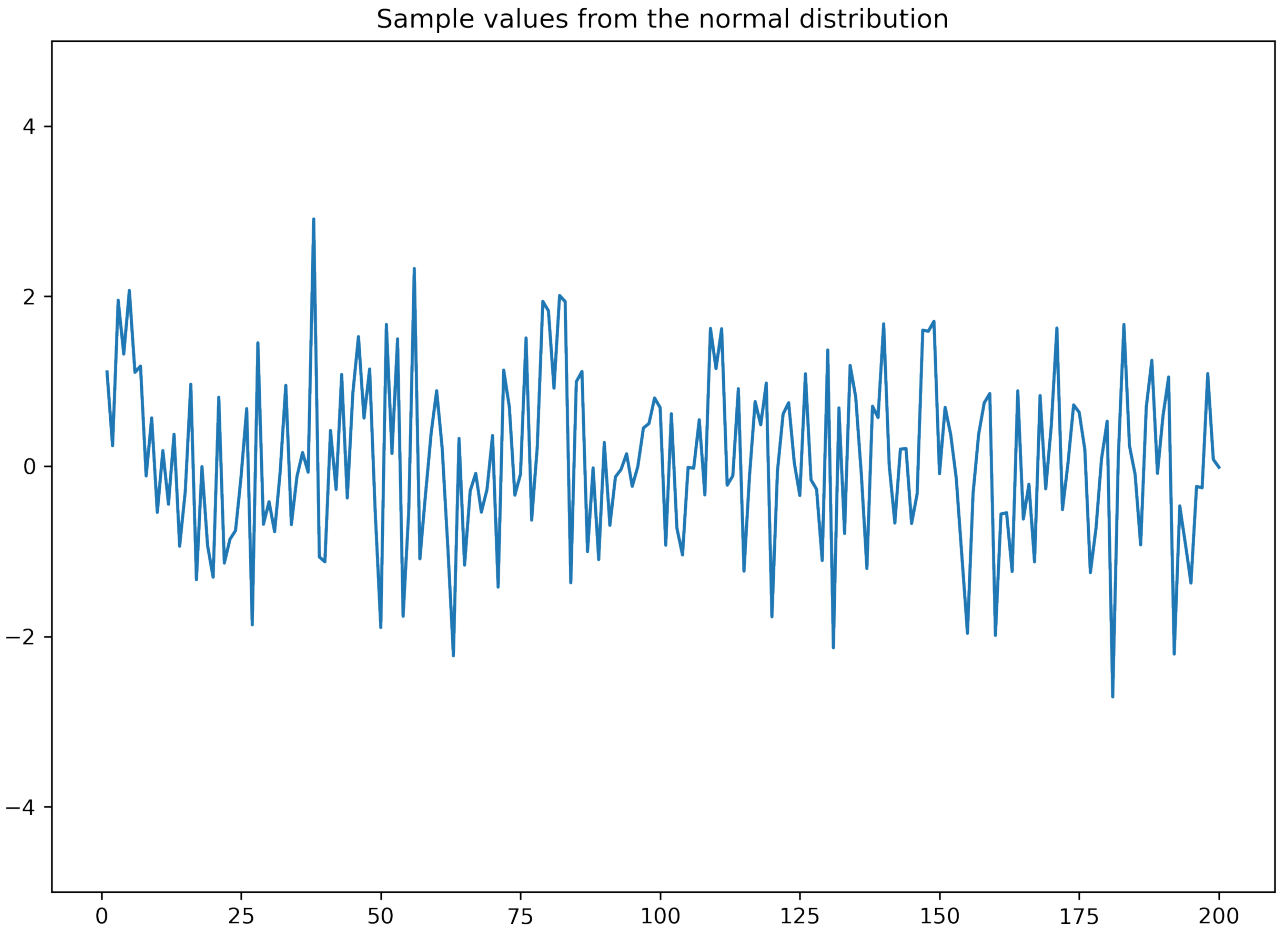
The normal distribution

- Has two parameters:
  - **Average value** μ = "center" of the function
  - **Standard deviation** σ = "width" of the function
    - Small σ = narrow and tall
    - Big σ = wide and low
- The front constant is just for normalization (ensures that integral $= 1$)
- Extremely often encountered in real life
- Any real value is possible ($w_A(x) > 0, \forall x \in \mathbb{R}$)
- Usually denoted as $\mathcal{N}(\mu, \sigma^2)$

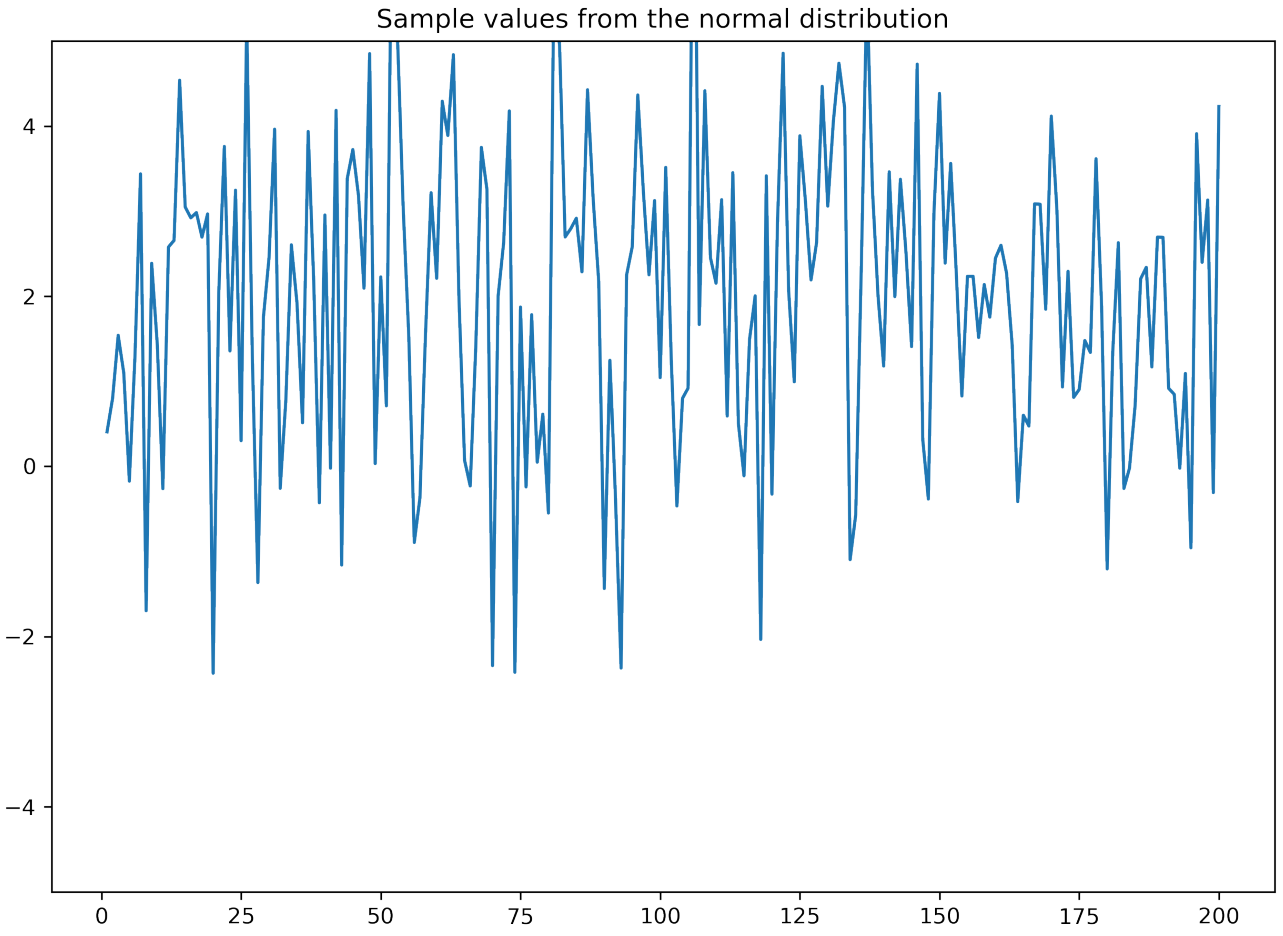
The normal distribution

- The distribution decreases as x gets farther from μ
  - Because of the term $-(x-\mu)^2$ at the exponent
  - Most likely values: around μ ($x - \mu = 0$)
  - Values closer to μ are more likely, values farther from μ are less likely
- The function describes a preference for values around μ, with decreasing preference when getting farther from μ

Example of values from the normal distribution (mu=0, sigma=1)



Example of values from the normal distribution (mu=2, sigma^2=4)



The empirical rule of normal distribution

For **any** normal distribution $\mathcal{N}(\mu, \sigma^2)$:

- About 68% of the values are within 1 standard deviation of the mean:

$$P\{\mu - \sigma \leq A \leq \mu + \sigma\} \approx 0.68$$

- About 95% of the values are within 2 standard deviations of the mean:

$$P\{\mu - 2\sigma \leq A \leq \mu + 2\sigma\} \approx 0.95$$

- About 99.7% of the values are within 3 standard deviations of the mean:

$$P\{\mu - 3\sigma \leq A \leq \mu + 3\sigma\} \approx 0.997$$

For a normal distribution, σ works like a "default unit of measure" for the distance from the mean

The empirical rule of normal distribution

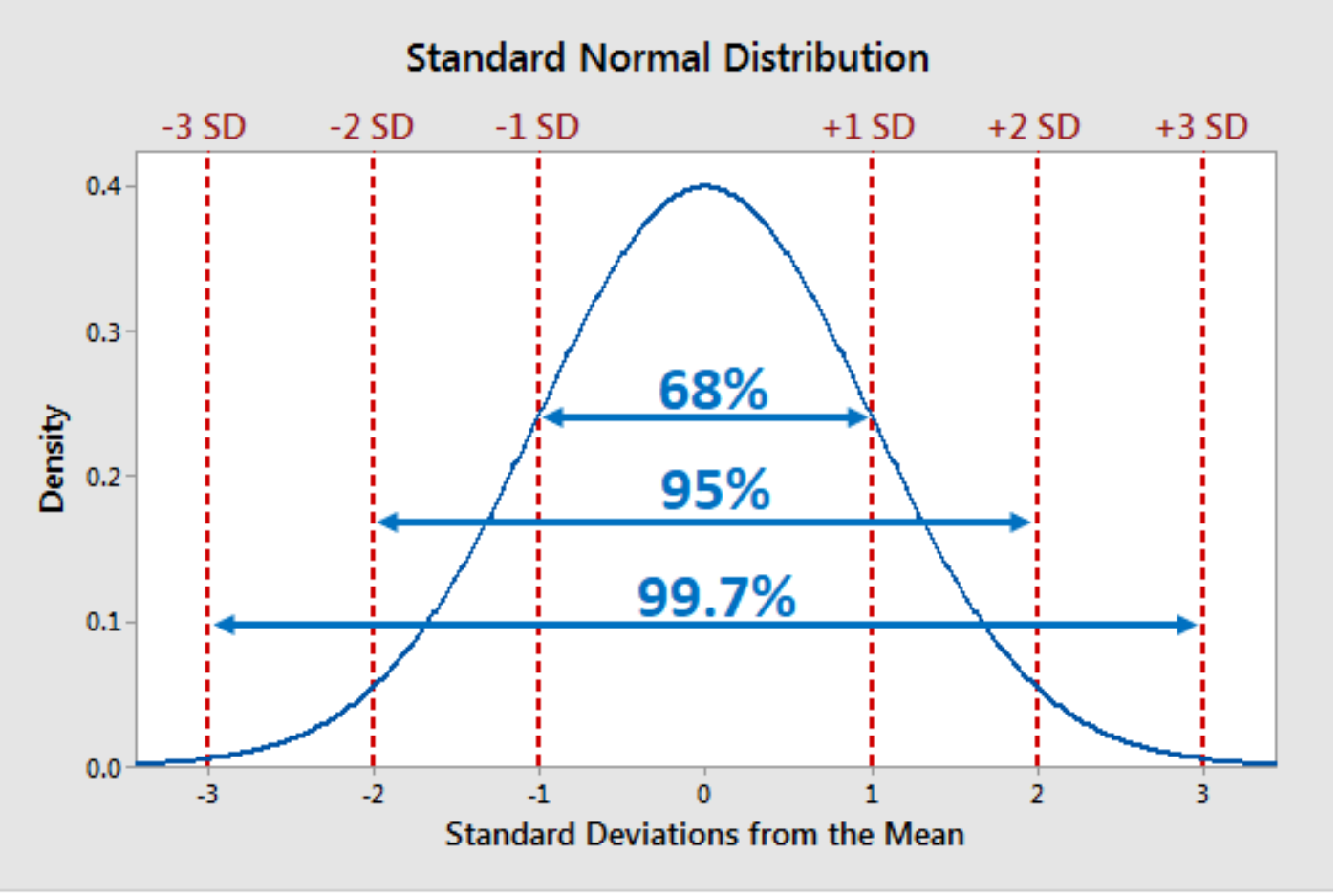
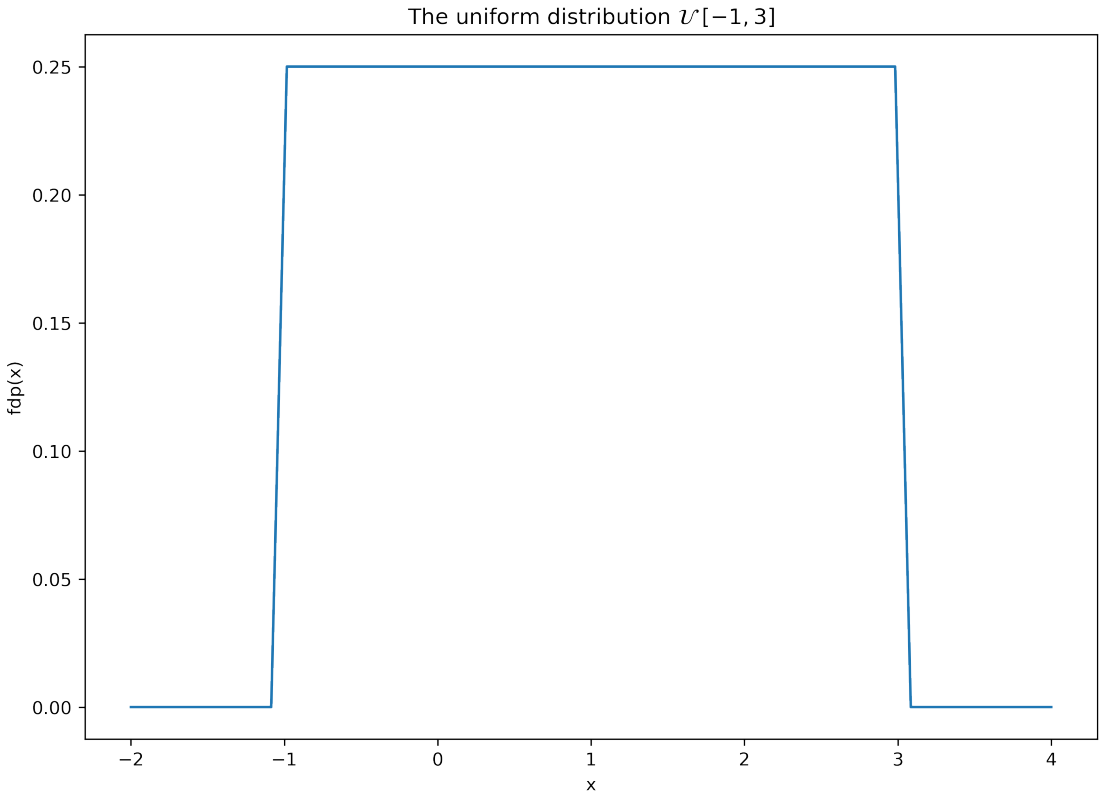


Figure 1: Empirical rule for any normal distribution

The uniform distribution

- The probability density function = a constant, between two endpoints

$$w_A(x) = \begin{cases} \frac{1}{b-a}, & x \in [a, b] \\ 0, & \textit{elsewhere} \end{cases}$$



The uniform distribution

- Has two parameters: the limits a and b of the interval
- The "height" of the function is $\frac{1}{b-a}$
  - in order for the integral to be 1
- Only values from the interval $[a, b]$ are possible
  - value cannot be outside interval (probability is 0)
- Denoted as $\mathcal{U}\ [a, b]$

Other distributions

- Many other distributions exist, relevant for particular applications

Computing probabilities for the normal distribution

- How to compute $\int_a^b$ for a normal distribution?
  - Can't be done with algebraic formula, non-elementary function
- Use *the error function*:

$$erf(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$$

- The CDF of a normal distribution $\mathcal{N}(\mu, \sigma^2)$

$$F_A(x) = \frac{1}{2}(1 + erf(\frac{x - \mu}{\sigma\sqrt{2}}))$$

- The values of *erf()* are available / are computed numerically
  - e.g. on Google, search for $erf(0.5)$
  - Other useful values:
    - $erf(-\infty) = -1$
    - $erf(\infty) = 1$

Exercise

Exercise:

- Let X be a r.v. with distribution $\mathcal{N}(3, 2)$. Compute the probability that $X \in [2, 4]$

Sum of constant + random variable

- Consider a random variable A
- What is $B = 5 + A$?

Answer:

- B is also a random variable
- B has same type of distribution, but the function is "shifted" by 5 to the right

Example:

- A is normal variable with $w_A(x) = \mathcal{N}(\mu = 3, \sigma^2 = 2)$
- What is the distribution of B $= 5 +$ A?
- Answer: $w_B(x) = \mathcal{N}(\mu = 8, \sigma^2 = 2)$

R.v. as functions of other r.v.

- A function applied to a r.v. produces another r.v.
- Examples: if A is a r.v. with distribution $\mathcal{U}\ [0, 10]$, then
  - $B = 5 + A$ is another r.v., with distribution $\mathcal{U}\ [5, 15]$
  - $C = A^2$ is also another r.v.
  - $D = cos(A)$ is also another r.v.
- Reason: since A is random, the values B, C, D are also random
- A, B, C, D are *not independent*
  - A certain value of one of them automatically implies the value of the others

Multiple random variables

- Consider a system with two continuous r.v. A and B
- What is the probability that the pair (A, B) has values around (x, y)?
- Distribution of the values of (A, B) is described by:
  - joint probability density function $w_{AB}(x, y)$
  - joint cumulative distribution function $F_{AB}(x, y)$

Multiple random variables

- Joint cumulative distribution function:

$$F_{AB}(x, y) = P\{A \leq x \cap B \leq y\}$$

- Joint probability density function:

$$w_{AB}(x, y) = \frac{\partial^2 F_{AB}(x, y)}{\partial x \partial y}$$

- The joint PDF gives the probability that the value of the pair (A, B) is in a vicinity of (x, y)
- Similar for discrete random variables

$$w_{AB}(x, y) = P\{A = x \cap B = y\}$$

Independent random variables

- Two r.v. A and B are **independent** if the value of one of them does not influence in any way the value of the other
- For independent r.v., the probability that A is around x and B is around y is **the product** of the two probabilities

$$w_{AB}(x, y) = w_A(x) \cdot w_B(y)$$

- Relation holds for CDF / PDF / PMF, continuous or discrete r.v.
- Same for more than two r.v.

Independent random variables

Exercise:

- Compute the probability that three r.v. X, Y and Z i.i.d. $\mathcal{N}(-1, 1)$ are all positive simultaneously
  - ***i.i.d*** $=$ "independent and identically distributed"

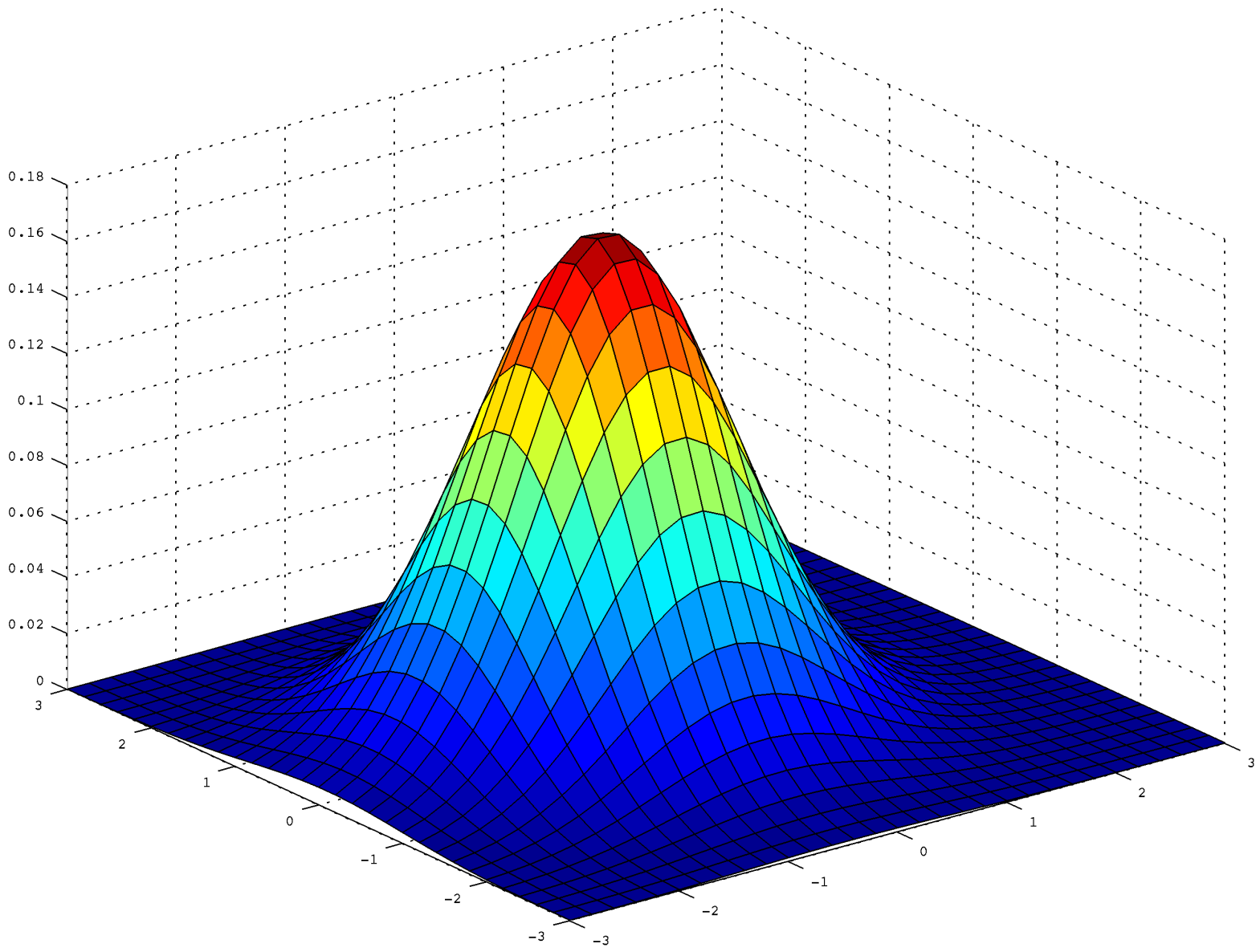
The 2D normal distribution (bivariate)

- The 2D normal distribution for a pair of random variables $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ is:

$$w_{\mathbf{x}}(\mathbf{x}) = \frac{1}{2\pi\sqrt{\det \Sigma}} e^{\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}$$

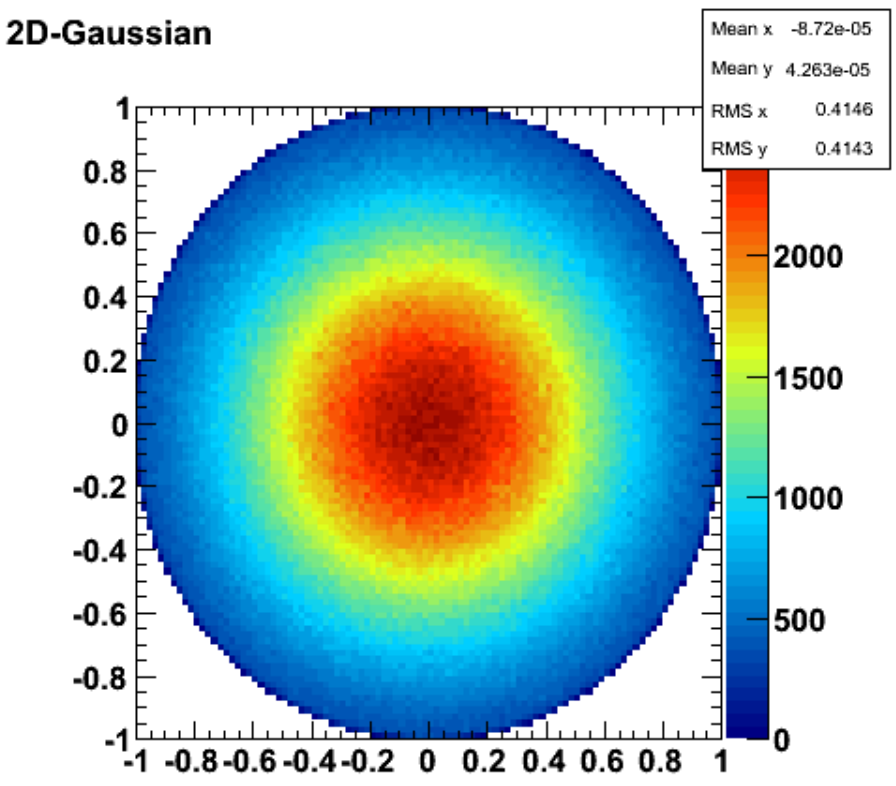
- It is characterized by a mean vector $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$, which sets the center, and a covariance matrix Σ with dimension 2×2 which sets the elliptical shape

2D normal distribution



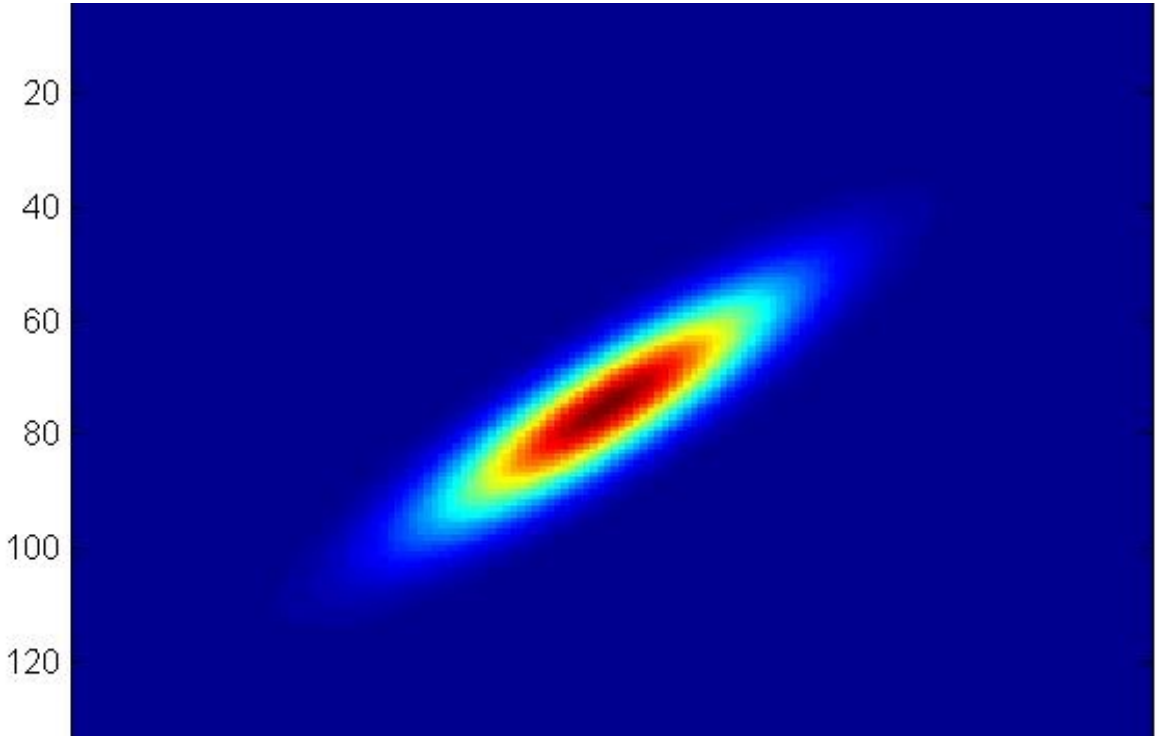
2D normal distribution - top view

- View from top
- Here, $\mu = (0, 0)$
- Probability density decreases as distance from center increases, in elliptical shapes (symmetrically)



Another 2D normal distribution

- View from top
- The ellipsoidal shape (orientation, elongation) is given by the covariance matrix



Example: car trajectories

Trajectories of cars in the Podu Ros intersection

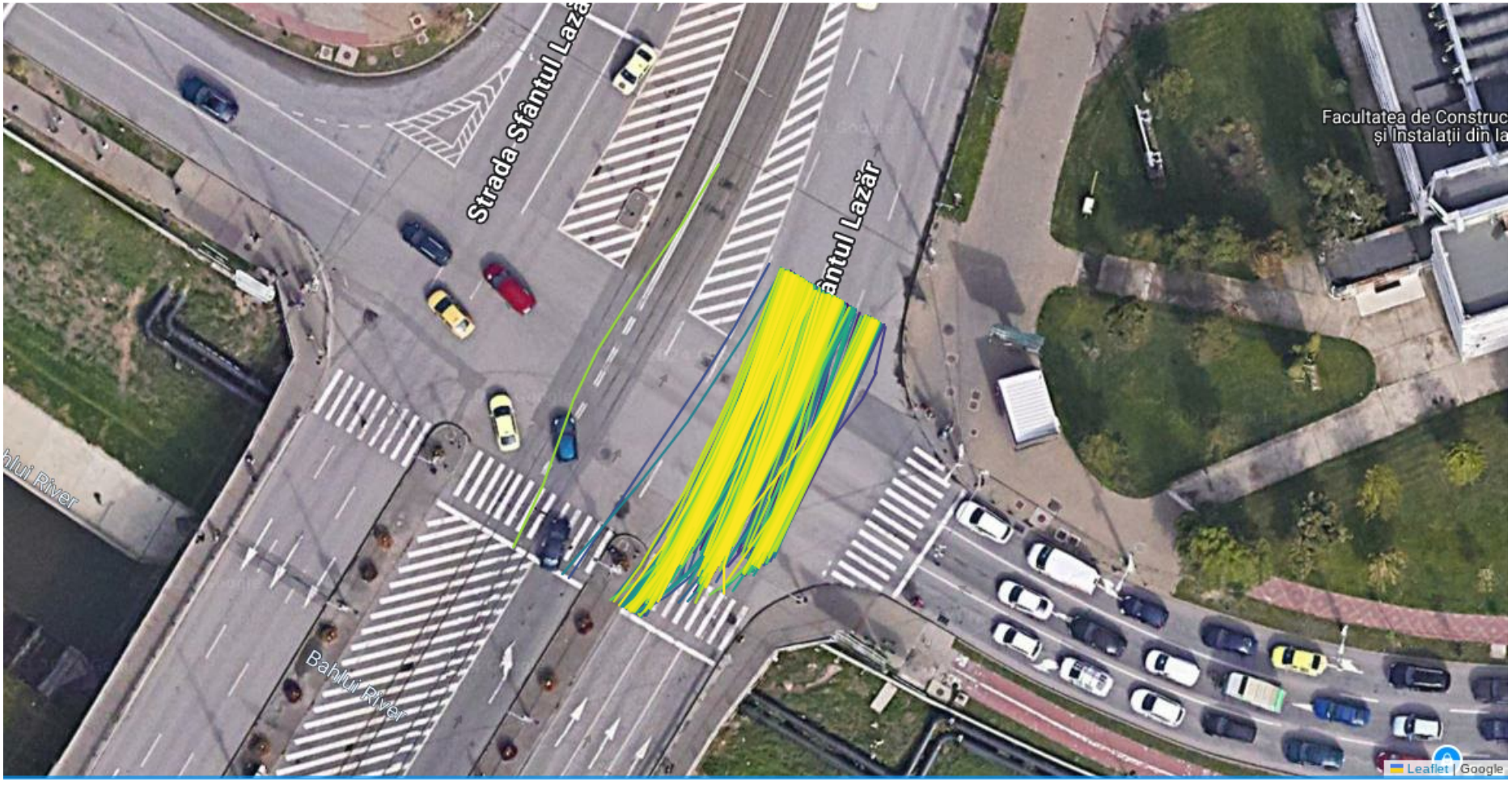
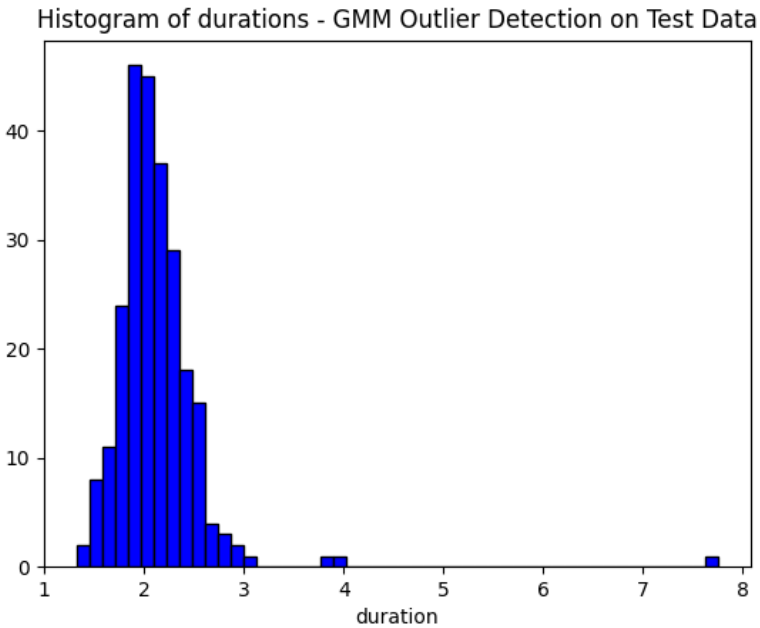


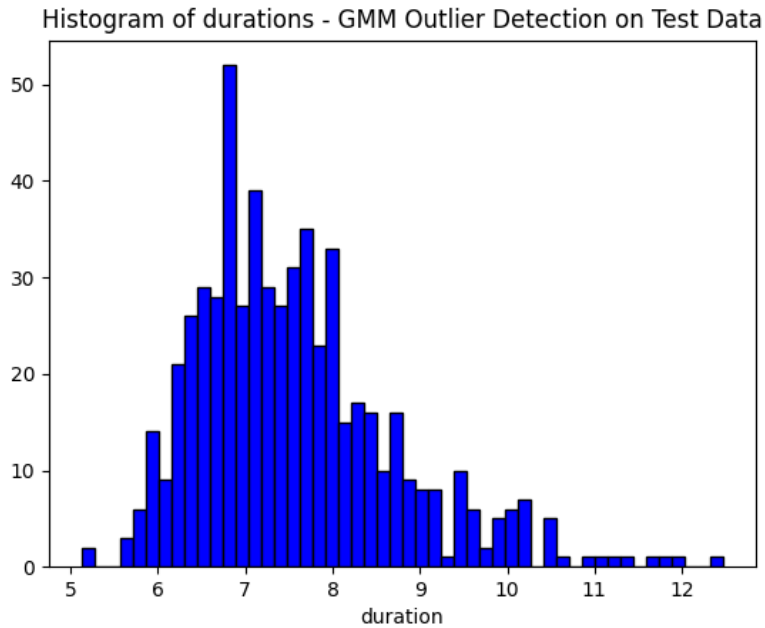
Figure 2: Trajectories of vehicles, In Podu Ros -> Out Palat

Example: car trajectories

Distribution of trajectory durations

- a. In Splai -> Out Palat
- b. In Splai -> Out Podu Ros

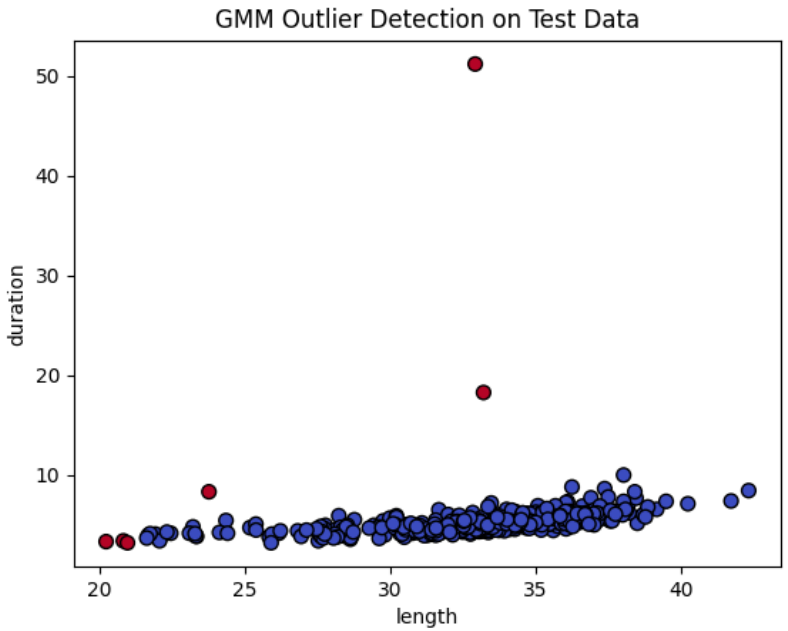


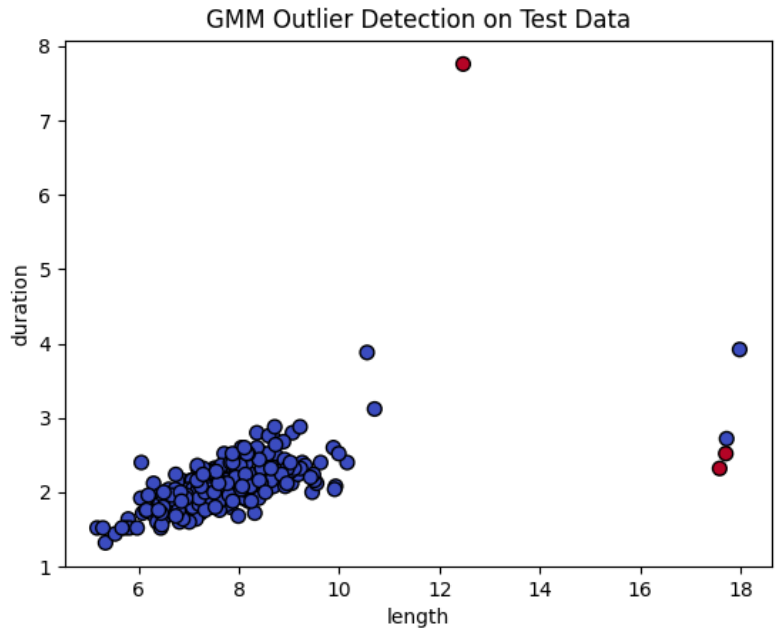


Example: car trajectories

Distribution of trajectory (length, duration)

- a. In Podu Ros -> Out Splai
- b. In Splai -> Out Podu Ros

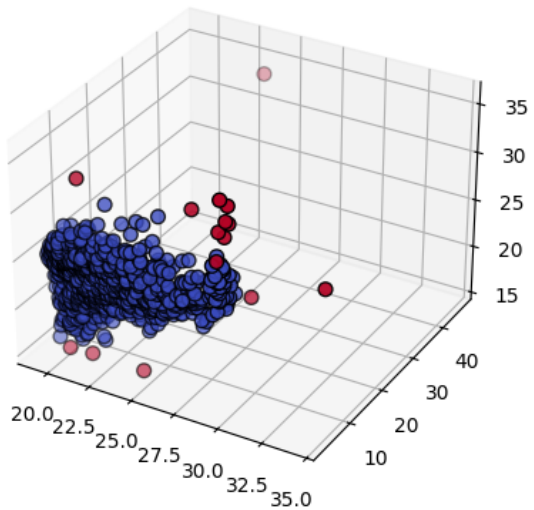


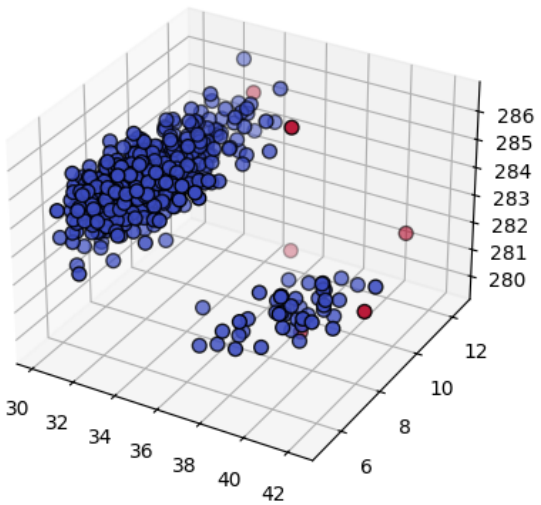


Example: car trajectories

Distribution of trajectory (length, duration, curvature)

a. In Podu Ros -> Out Palat
b. In Splai -> Out Podu Ros





Statistical averages

- R.v. are described by statistical averages ("*moments*")
- **The average value** (or "expected value", or "moment of order 1")
- Continuous r.v.:

$$\overline{A} = E\{A\} = \int_{-\infty}^{\infty} x \cdot w_A(x) dx$$

- Discrete r.v.:

$$\overline{A} = E\{A\} = \sum_{x=-\infty}^{\infty} x \cdot w_A(x)$$

- (Example: the entropy of H(X) = the average value of the information)
- Usual notation: μ

Meaning of average value

- What does the average value of a random variable *mean*?
  - If we have $N \to \infty$ values drawn from this distribution, it is their average;
  - If you'd have to guess the value of a random variable value X, and you pay a price proportional to square of the error of your guess u, $(u - X)^2$, the average value μ would be the best choice, as it minimizes the overall cost:

$$\mu = \arg\min_u \int_{-\infty}^{\infty} (u - x)^2 \cdot w(x) dx$$

  - Proof: on blackboard: derivate and set derivative to 0

Meaning of average value

- Values with high probability will "pull" the average value closer to them
- For distributions with symmetrical shape (like the normal distribution), the average value is always the middle value
  - Proof: both sides of the function "pull" the average value equally, so it remains in the middle
- For the normal distribution, $\overline{X} = \mu$ (obvious from notation)
- For the uniform distribution $\mathcal{U}[a, b]$, $\overline{X} = \frac{a+b}{2}$ (middle)

Properties of the average value

- Computing the average value is a **linear** operation
  - because the underlying integral / sum is a linear operation
- If A and B are two random variables:
- Linearity

$$E\{c_1 A + c_2 B\} = c_1 E\{A\} + c_2 E\{B\}$$

- Or:

$$E\{cA\} = cE\{A\}, \forall c \in \mathbb{R}$$

$$E\{A + B\} = E\{A\} + E\{B\}$$

- No proof given here

Average squared value

- **Average squared value** = average value of the squared values
- "Moment of order 2"
- Continuous r.v.:

$$\overline{A^2} = E\{A^2\} = \int_{-\infty}^{\infty} x^2 \cdot w_A(x) dx$$

- Discrete r.v.:

$$\overline{A^2} = E\{A^2\} = \sum_{-\infty}^{\infty} x^2 \cdot w_A(x)$$

- Interpretation: average of squared values = average power of a signal

Variance

- **Variance**= average squared value of the difference to the average value
- Continuous r.v.:

$$\sigma^2 = \overline{\{A-\mu\}^2} = \int_{-\infty}^{\infty} (x-\mu)^2 \cdot w_A(x) dx$$

- Discrete r.v.:

$$\sigma^2 = \overline{\{A-\mu\}^2} = \sum_{-\infty}^{\infty} (x-\mu)^2 \cdot w_A(x)$$

- Interpretation: how much do the values vary around the average value
  - σ^2 = large: large spread around the average value
  - σ^2 = small: values are concentrated around the average value

Relation between the three values

- Relation between the average value, the average squared value, and the variance:

$$\begin{aligned}\sigma^2 &= \overline{\{A-\mu\}^2} \\ &= \overline{A^2 - 2\cdot A\cdot \mu + \mu^2} \\ &= \overline{A^2} - 2\mu\overline{A} + \mu^2 \\ &= \overline{A^2} - \mu^2\end{aligned}$$

Sum of random variables

- Sum of two or more r.v. is also a r.v.
- If the variables are **independent**, the distribution of their sum = the **convolution** of the distributions of the two r.v.
- If $C = A + B$

$$w_C(x) = w_A(x) \star w_B(x)$$

- Particular case: if A and B are normal r.v., with $\mathcal{N}(\mu_A, \sigma_A^2)$ and $\mathcal{N}(\mu_B, \sigma_B^2)$, then:
  - C is also a normal r.v., with $\mathcal{N}(\mu_C, \sigma_C^2)$, having:
  - average = sum of the two averages: $\mu_C = \mu_A + \mu_B$
  - variance = sum of the two variances: $\sigma_C^2 = \sigma_A^2 + \sigma_B^2$

I.2 Random processes

Random process

- A **random process** = a sequence of random variables indexed in time
- **Discrete-time** random process $f[n]$ = a sequence of random variables at discrete moments of time
  - e.g.: a sequence 50 of throws of a dice, the daily price on the stock market
- **Continuous-time** random process $f(t)$ = a continuous sequence of random variables at every moment
  - e.g.: a noise voltage signal, a speech signal
- Every sample from a random process is a (different) random variable!
  - e.g. $f(t_0)$ = value at time t_0 is a r.v.

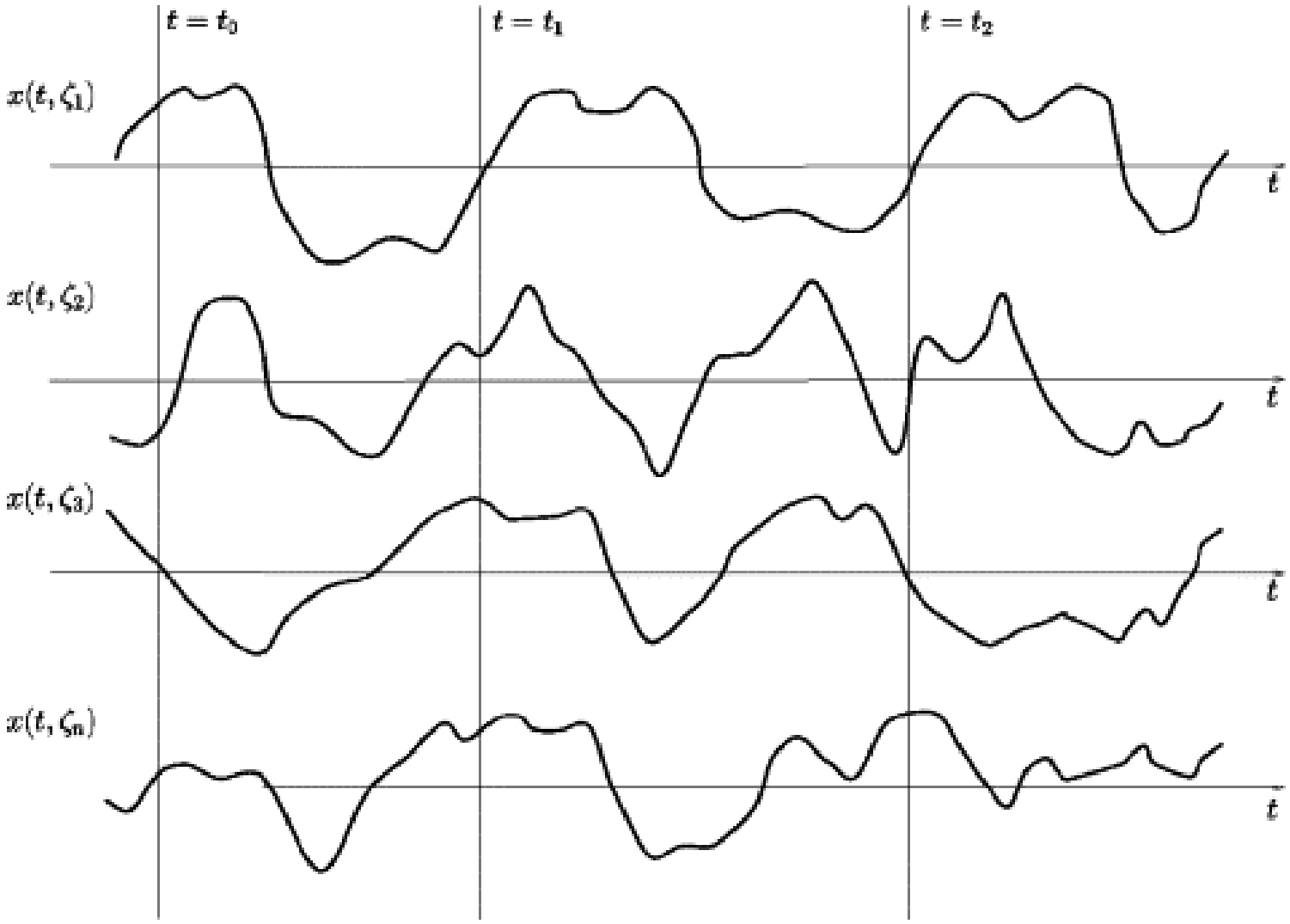
Realizations of random processes

- A **realization** of the random process = a particular sequence of realizations of the underlying r.v.
  - e.g. we see a given noise signal on the oscilloscope, but *we could have seen any other realization just as well*
- Typically denoted as $f^{(k)}[n]$ or $f^{(k)}(t)$
  - k indicates the particular realization that we consider
- When we consider a random process = we consider the set of all possible realizations

Random process is 2-D

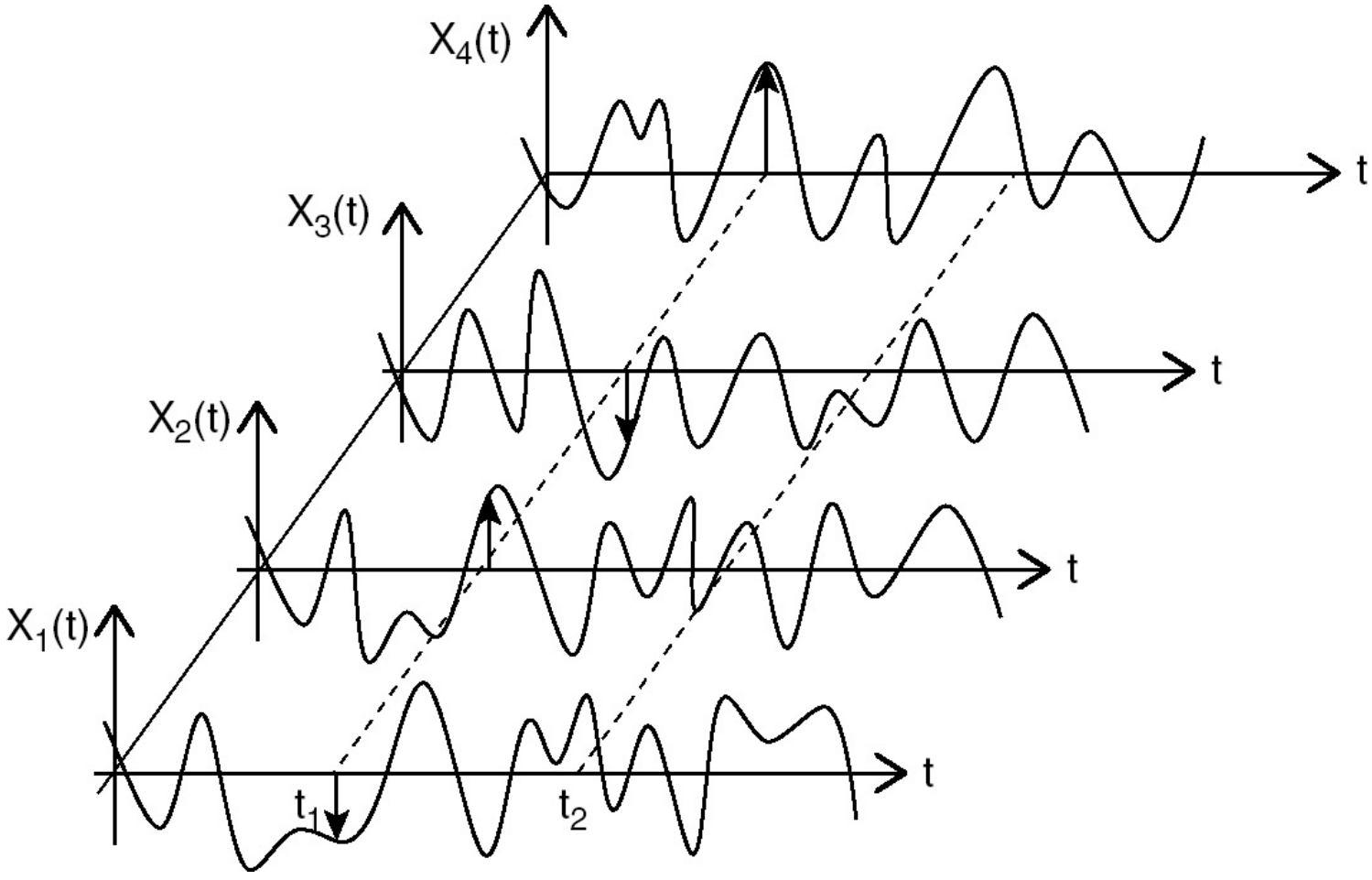
- A random process is a 2-Dimensional thing
  - $f^{(k)}[n]$ or $f^{(k)}(t)$ depends on two variables:
    - k = the particular realization
    - t or n = time

Random process is 2-D



- from "Information-Based Inversion and Processing with Applications" Edited by Tadeusz J. Ulrych, Mauricio D. Sacchi, Volume 36,

Random process is 2-D



- from: Razdolsky, L. (2014). Random Processes. In Probability-Based Structural Fire Load (pp. 89-136). Cambridge: Cambridge University Press

Random process is 2-D

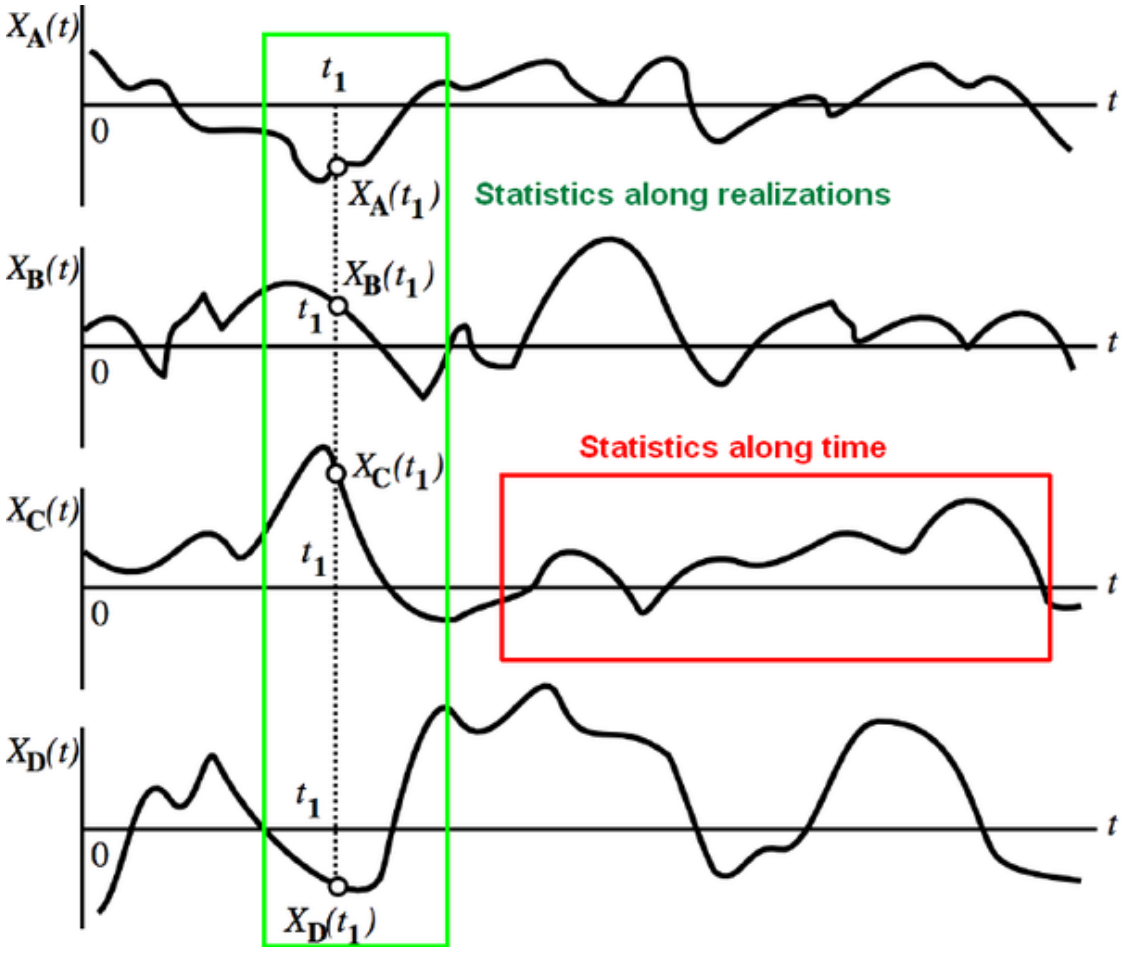


- from: https://www.quora.com/What-is-the-difference-between-a-stationary-ergodic-and-a-stationary-non-ergodic-process

Two types of averages

- Random processes have two types of averages:
  - **Statistical** averages = for a given time t or n, across all possible realizations
  - **Temporal** averages = for a given realization k, across all time

Two types of averages



- from: https://www.quora.com/What-is-the-difference-between-a-stationary-ergodic-and-a-stationary-non-ergodic-process

Distributions of order 1 of random processes

- Every sample $f(t_1)$ from a random process is a random variable
  - it is described by a **distribution of order 1**
  - has a CDF $F_1(x; t_1)$
  - has a PDF $w_1(x; t_1) = \frac{dF_1(x;t_1)}{dx}$
  - everything depends on the time moment t_1
- The sample at time t_2 is a different random variable with **possibly different** functions
  - has a different CDF $F_1(x; t_2)$
  - has a different PDF $w_1(x; t_2) = \frac{dF_1(x;t_2)}{dx}$
- These functions specify how the value of one sample is distributed
- The index w_1 indicates we consider a single random variable (distribution of order 1)
- Same for discrete-time random processes

Distributions of order 2

- A pair of random variables $f(t_1)$ and $f(t_2)$ form a system of 2 r.v.
  - they are described by a **distribution of order 2**
  - have a joint CDF $F_2(x_i, x_j; t_1, t_2)$
  - have a joint PDF $w_2(x_i, x_j; t_1, t_2) = \frac{\partial^2 F_2(x_i, x_j; t_1, t_2)}{\partial x_i \partial x_j}$
  - depend on time moments t_1 and t_2
- These functions specify how the pair of values is distributed
- Same for discrete-time random processes

Distributions of order n

- Generalize to n samples of the random process
- A set of n random variables $f(t_1), \ldots f(t_n)$ from the random process $f(t)$
  - are described by **distribution of order n**
  - have joint CDF $F_n(x_1, \ldots x_n; t_1, \ldots t_n)$
  - have joint PDF $w_n(x_1, \ldots x_n; t_1, \ldots t_n) = \frac{\partial^n F_n(x_1, \ldots x_n; t_1, \ldots t_n)}{\partial x_1 \ldots \partial x_n}$
  - depend on time moments $t_1,\ t_2,\ \ldots\ \ t_n$
- These functions specify how the whole set of n values is distributed
- Same for discrete-time random processes

Statistical averages

Random processes are characterized using **statistical** and **temporal** averages ("moments")

For continuous random processes:

1. **Average value**

$$\overline{f(t_1)} = \mu(t_1) = \int_{-\infty}^{\infty} x \cdot w_1(x; t_1) dx$$

2. **Average squared value** ("valoarea pătratică medie")

$$\overline{f^2(t_1)} = \int_{-\infty}^{\infty} x^2 \cdot w_1(x; t_1) dx$$

Statistical averages - variance

3. **Variance** ("varianța")

$$\sigma^2(t_1) = \overline{\{f(t_1) - \mu(t_1)\}^2} = \int_{-\infty}^{\infty} (x - \mu(t_1))^2 \cdot w_1(x; t_1) dx$$

- Relation between these three:

$$\begin{aligned}\sigma^2(t_1) =& \overline{\{f(t_1) - \mu(t_1)\}^2} \\ =& \overline{f(t_1)^2 - 2f(t_1)\mu(t_1) + \mu(t_1)^2} \\ =& \overline{f^2(t_1)} - \mu(t_1)^2\end{aligned}$$

- Note:
  - these three values are calculated across all realizations, at time t_1
  - they characterize only the sample at time t_1
  - at a different time t_2, the r.v. $f(t_2)$ is different so all average values might be different

Statistical averages - autocorrelation

Statistical averages for a pair of **two samples**:

4. **The autocorrelation function**

$$R_{ff}(t_1, t_2) = \overline{f(t_1)f(t_2)} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 x_2 w_2(x_1, x_2; t_1, t_2) dx_1 dx_2$$

5. **The correlation function** (for different random processes $f(t)$ and $g(t)$)

$$R_{fg}(t_1, t_2) = \overline{f(t_1)g(t_2)} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x_1 y_2 w_2(x_1, y_2; t_1, t_2) dx_1 dy_2$$

- Note:
  - these functions may have different values for a different pair of samples at times (t_1, t_2)

Discrete values

If the values are discrete: integral becomes sum, notation $f(t)$ becomes $f[t]$):

1. $\overline{f[t_1]} = \mu(t_1) = \sum_{x=-\infty}^{\infty} x \cdot w_1(x; t_1)$
2. $\overline{f^2[t_1]} = \sum_{x=-\infty}^{\infty} x^2 \cdot w_1(x; t_1)$
3. $\sigma^2(t_1) = \overline{\{f[t_1] - \mu(t_1)\}^2} = \sum_{x=-\infty}^{\infty} (x - \mu(t_1))^2 \cdot w_1(x; t_1)$
4. $R_{ff}(t_1, t_2) = \overline{f[t_1] f[t_2]} = \sum_{x_1=-\infty}^{\infty} \sum_{x_2=-\infty}^{\infty} x_1 x_2 w_2(x_1, x_2; t_1, t_2)$
5. $R_{fg}(t_1, t_2) = \overline{f[t_1] g[t_2]} = \sum_{x_1=-\infty}^{\infty} \sum_{y_2=-\infty}^{\infty} x_1 y_2 w_2(x_1, y_2; t_1, t_2)$

Temporal averages

- What to do when we only have access to a single realization $f^{(k)}(t)$?
- Compute values **for a single realization $f^{(k)}(t)$, across all time moments**

Temporal averages

Temporal averages for continuous random processes:

1. **Temporal average value**

$$\overline{f^{(k)}(t)} = \mu^{(k)} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} f^{(k)}(t)dt$$

2. **Temporal average squared value**

$$\overline{[f^{(k)}(t)]^2} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} [f^{(k)}(t)]^2 dt$$

Temporal variance

3. **Temporal variance**

$$\sigma^2 = \overline{\{f^{(k)}(t) - \mu^{(k)}\}^2} = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} (f^{(k)}(t) - \mu^{(k)})^2 dt$$

- Relation between the three:

$$\sigma^2 = \overline{[f^{(k)}(t)]^2} - [\mu^{(k)}]^2$$

- Note:
  - these values do not depend anymore on time t (integrated)

Temporal autocorrelation

4. The **temporal autocorrelation function**

$$\begin{aligned} R_{ff}(t_1, t_2) &= \overline{f^{(k)}(t_1 + t) f^{(k)}(t_2 + t)} \\ &= \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} f^{(k)}(t_1 + t) f^{(k)}(t_2 + t) dt \end{aligned}$$

5. The **temporal correlation function** (for different random processes $f(t)$ and $g(t)$)

$$\begin{aligned} R_{fg}(t_1, t_2) &= \overline{f^{(k)}(t_1 + t) g^{(k)}(t_2 + t)} \\ &= \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} f^{(k)}(t_1 + t) g^{(k)}(t_2 + t) dt \end{aligned}$$

Discrete random processes

For **discrete random processes**: replace $\int$ with $\sum$, T with N, and divide to $2N+1$ instead of $2T$

1. $\overline{f^{(k)}[t]} = \mu^{(k)} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{t=-N}^{N} f^{(k)}[t]$
2. $\overline{[f^{(k)}[t]]^2} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{t=-N}^{N} (f^{(k)}[t])^2$
3. $\sigma^2 = \overline{\left\{f^{(k)}[t] - \mu^{(k)}\right\}^2} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{t=-N}^{N} (f^{(k)}[t] - \mu^{(k)})^2$

Discrete random processes

4. Temporal autocorrelation:

$$\begin{aligned} R_{ff}(t_1, t_2) &= \overline{f^{(k)}[t_1 + t] f^{(k)}[t_2 + t]} \\ &= \lim_{N \to \infty} \frac{1}{2N + 1} \sum_{t=-N}^{N} f^{(k)}[t_1 + t] f^{(k)}[t_2 + t] \end{aligned}$$

5. Temporal correlation:

$$\begin{aligned} R_{fg}(t_1, t_2) &= \overline{f^{(k)}[t_1 + t] g^{(k)}[t_2 + t]} \\ &= \lim_{N \to \infty} \frac{1}{2N + 1} \sum_{t=-N}^{N} f^{(k)}[t_1 + t] g^{(k)}[t_2 + t] \end{aligned}$$

Finite length realizations

- We may have a realization of **finite length** (e.g. a vector with 1000 elements)
- How do we compute the temporal averages?
- Just use $\int_{t_{min}}^{t_{max}}$ or $\sum_{t_{min}}^{t_{max}}$ instead of $-\infty$ to ∞
- Example: Compute the temporal averages (all five) for the finite-length realization:

$$\{1, -1, 2, -2, 3, -3, 4, -4, 5, -5\}$$

Statistical and temporal averages

- Statistical averages are usually the ones we want
  - but they require to know the distributions $w(x)$, which is most often unknown
- In real life, typically we can only measure one realization
  - so we can only compute the temporal values for one realization
- Fortunately, in many cases statistical and temporal averages **are the same** (ergodicity)

Stationary random processes

- Until now, all the statistical averages are dependent on time
  - i.e. they might be different for a sample at t_1 and another at t_2
- **Stationary** random process = when all statistical averages are identical if we shift the time origin (e.g. delay the signal)
- Equivalent definition: if all the PDF are identical when shifting the time origin

$$w_n(x_1, ...x_n; t_1, ...t_n) = w_n(x_1, ...x_n; t_1 + \tau, ...t_n + \tau)$$

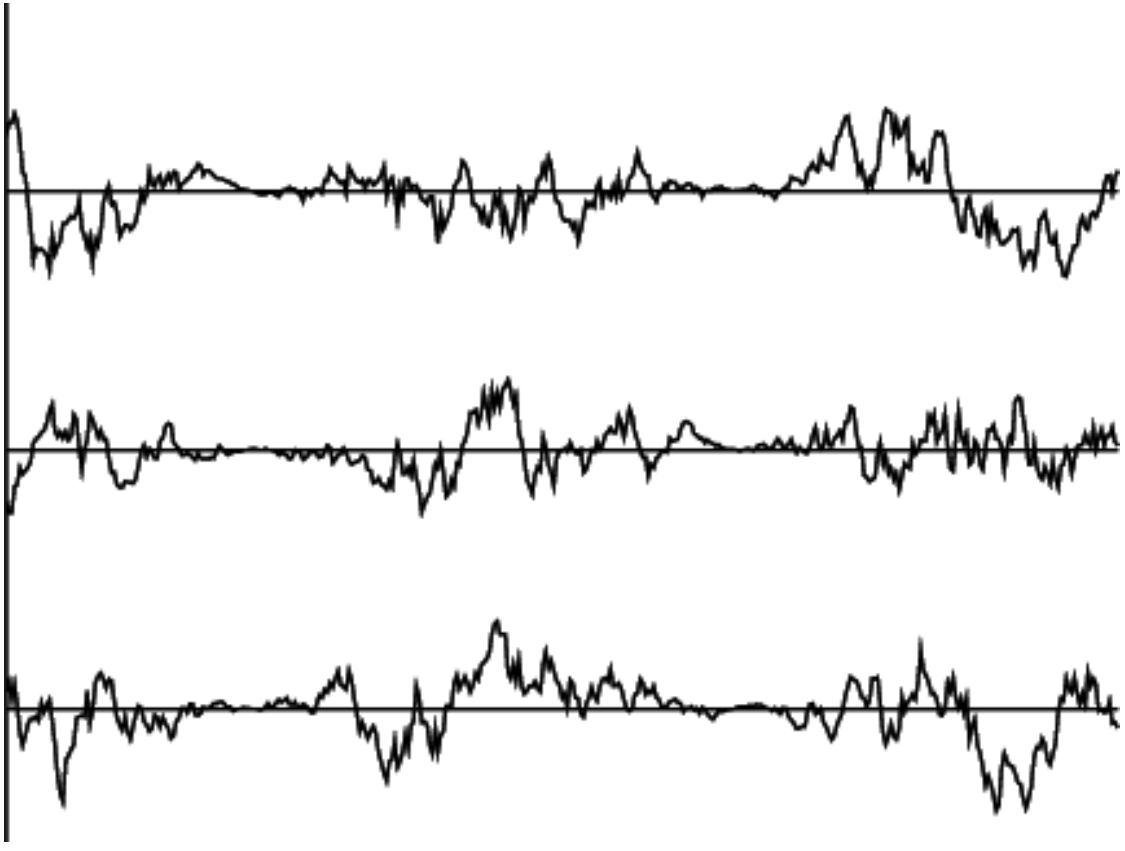
- Basically, stationary = **all statistical averages do not depend on time** t

Strict-sense and wide-sense stationary

- **Strictly** stationary / **strongly** stationary / **strict-sense** stationary:
  - relation holds for every n (i.e. for distributions of any order)
  - average value, average squared value, variance, autocorrelation, **and all other** higher-order statistics do not depend on origin of time t
- **Weakly** stationary / **wide-sense** stationary:
  - relation holds only for $n = 1$ and $n = 2$ (i.e. for distributions of a single sample, and of a pair of samples)
  - **only** the average value, average squared value, variance, autocorrelation do not depend on time t, but high-order statistics may depend

Stationary random processes

- Is the Random Process below stationary or non-stationary?



- from: SEX, LIES & STATISTICS, Ned Wright, http://www.astro.ucla.edu/~wright/statistics/

Stationary random processes

- Answer: non-stationary
- You can see that the variance is not the same at all moments of time

Consequences of stationarity

- For distributions of a single sample (order $n = 1$):

$$w_1(x_i; t_1) = w_1(x_i; t_2) = w_1(x_i)$$

- The average value, average squared value, variance of a sample are all **identical at any time** t

$$\overline{f(t)} = constant = \overline{f}, \forall t$$

$$\overline{f^2(t)} = constant = \overline{f^2}, \forall t$$

$$\sigma^2(t) = constant = \sigma^2, \forall t$$

Consequences of stationarity

- For distributions of a pair of samples (order $n = 2$):

$$w_2(x_i, x_j; t_1, t_2) = w_2(x_i, x_j; 0, t_2 - t_1) = w_2(x_i, x_j; t_2 - t_1)$$

- The autocorrelation & correlation function depend only on the **time difference** $\tau = t_2 - t_1$ between the samples

$$R_{ff}(t_1, t_2) = R_{ff}(0, t_2 - t_1) = R_{ff}(\tau) = \overline{f(t)f(t+\tau)}$$

- Depend on a single value $\tau = t_2 - t_1 =$ time difference of the two samples

Consequences of stationarity

Definition of autocorrelation function for a **stationary** r.p:

- Statistical autocorrelation: no change
- Temporal autocorrelation:
  - for continuous r.p.

$$R_{ff}(\tau) = \overline{f(t)f(t+\tau)} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} f^{(k)}(t) f^{(k)}(t+\tau) dt$$

  - for discrete r.p.

$$R_{ff}(\tau) = \overline{f(t)f(t+\tau)} = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{t=-N}^{N} f^{(k)}[t] f^{(k)}[t+\tau]$$

  - finite length realizations: use $\int_{t_{min}}^{t_{max}}$ or $\sum_{t_{min}}^{t_{max}}$

Consequences of stationarity

- For **correlation**, similar to the autocorrelation definition on the previous slide
- Correlation depends only on the **time difference** $\tau = t_2 - t_1$ between the samples

$$R_{fg}(t_1, t_2) = R_{fg}(0, t_2 - t_1) = R_{fg}(\tau) = \overline{f(t)g(t+\tau)}$$

Interpretation of autocorrelation

- $R_{ff}(\tau) = \overline{f(t)f(t+\tau)}$ = the **average value of the product of two samples which are time τ apart**
  - e.g. tells us if the two samples vary in same direction or not
- Same for correlation, but the samples are taken from different r.p f and g

Interpretation of autocorrelation

- Example:
  - $R_{ff}(0.5) > 0$ means two samples separated by 0.5 seconds tend to vary in same direction (both positive, both negative $=>$ their product is mostly positive)
    - knowing one of them, you can "guess" the other one
  - $R_{ff}(1) < 0$ means two samples separated by 1 second tend to vary in opposite directions (when one is positive, the other is negative $=>$ their product is mostly negative)
    - knowing one of them, you can "guess" the other one
  - $R_{ff}(2) = 0$ means two samples separated by 2 seconds are **uncorrelated** (their product is 0 on average, so the two samples have equal chances of being in the same or opposite directions)
    - knowing one of them, you **can't** "guess" the other one

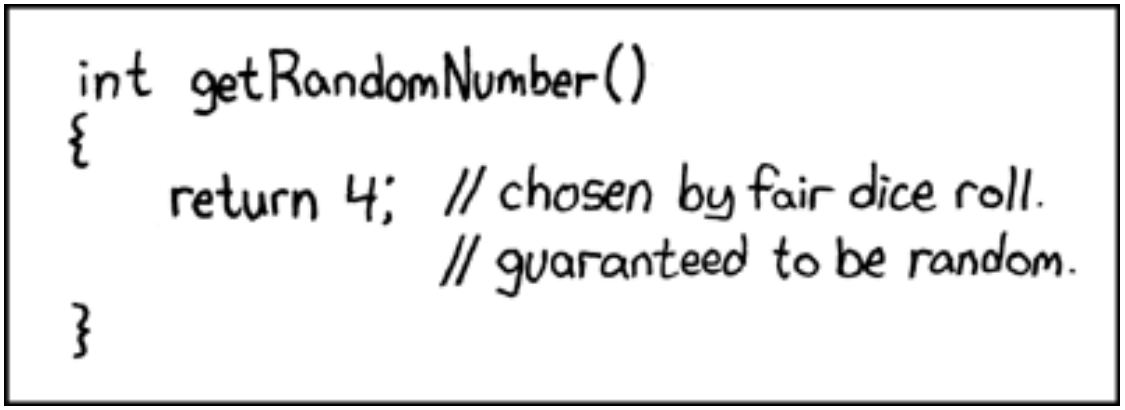
Ergodic random processes

- In practice, we have access to a single realization
- **Ergodic** random process = if the temporal averages on any realization are **equal to the statistical** averages
- Ergodicity means:
  - We can compute / estimate all averages from a single realization (any)
    - but the realization must be very long (length $\to \infty$) for precise results
  - Realizations are all similar to the others, in terms of averages
    - so a single realization is characteristic of the whole process

Ergodic random processes

- Most random processes we consider in this course are both ergodic and stationary
  - e.g. voltage noises
- Example of **non-ergodic** process:
  - throw a dice, then the next 50 values are identical to the first
    - a single realization is not characteristic

Ergodic random processes



- XKCD 221 (link here: https://xkcd.com/221/)
- Consider all numbers which could have been obtained by the "fair dice roll"
- What's the problem here?
  - stationary or non-stationary?
  - ergodic or non-ergodic?

I.3 More on autocorrelation

The Power Spectral Density of a signal

- The **Power Spectral Density (PSD)** $S_{ff}(\omega) =$ a function indicating the power of the signal depending on the frequency f $(\omega = 2\pi f)$
- For a **deterministic** signal (i.e. non-random), it is given by the modulus of the Fourier transform (squared):

$$S_{ff}(\omega) = |F(\omega)|^2$$

- The power in the frequency band $[f_1, f_2]$ is equal to $\frac{1}{2\pi} \cdot \int_{2\pi f_1}^{2\pi f_2} S_{ff}(\omega) d\omega$
- The whole power of the signal is $P = \frac{1}{2\pi} \cdot \int_{-\infty}^{\infty} S_{ff}(\omega) d\omega$
- The PSD is a measurable quantity
  - it can be determined experimentally
  - it is important in practical (engineering) applications

The Power Spectral Density of a random process

- How about for a random process?
  - we don't have one signal, we have an infinite number of possible realizations
  - we have a (different) Fourier transform for each possible realization
  - the PSD for every realization is, itself, a random process
- The **PSD of a random process** = the average value of the PSD for all the realizations
- Has the same meaning and use as for a deterministic signal, but **in average** across all possible realizations
  - for a particular realization, the values vary around the average

The Wiener-Khinchin theorem

The Wiener-Khinchin theorem ("teorema Wiener-Hincin")

- The Power Spectral Density of a random process = **the Fourier transform of the autocorrelation function**

$$S_{ff}(\omega) = \int_{-\infty}^{\infty} R_{ff}(\tau) e^{-j\omega\tau} d\tau$$

$$R_{ff}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{ff}(\omega) e^{j\omega\tau} d\omega$$

- No proof
- Notes:
  - Makes a relation between two rather different domains:

1. The autocorrelation function: a *statistical* property
2. The PSD function: a *physical* property (relevant for engineering purposes)

White noise

- **White noise** = a random process with **autocorrelation function equal to a Dirac** function

$$R_{ff}(\tau) = A \cdot \delta(\tau)$$

  - is a random process: every sample of white noise is a random variable
  - autocorrelation is a Dirac: autocorrelation is 0 for any $\tau \neq 0$
  - any two different samples ($\tau \neq 0$) have zero correlation (are uncorrelated)
    - values of any two different samples are not related

White noise

- Power spectral density of white noise = Fourier transform of a Dirac = **a constant** = does not depend on ω

$$S_{ff}(\omega) = constant, \forall \omega$$

  - equal distribution of power at all frequencies up to ∞
- White noise can have **any distribution** (normal, uniform etc.)
  - the term "white noise" doesn't refer to the distribution of sample values, but to the fact that all samples are unrelated to each other

Band-limited white noise

- In real life, power spectral density goes to 0 at very high frequencies
  - because total power $P = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{ff} \omega d\omega$ cannot be infinite
  - known as "**band-limited white noise**"
- In this case, autocorrelation = **approximately** a Dirac, but not infinitely thin
  - samples which are very close are necessarily a bit correlated
  - e.g. due to small parasitic capacities

AWGN

- **AWGN** = Additive White Gaussian Noise
  - is the usual type of noise considered in applications
- It means:
  - additive: the noise is added to the original signal (e.g. not multiplied with it)
  - white: the samples are uncorrelated (unrelated) with each other
  - gaussian: the samples have normal distribution
  - noise: it is a random process, every sample is random, every realization is different

2020-2021 Exam

- Chapter 1 ends here for 2020-2021 exam. Following slides not needed.

Properties of the autocorrelation function

1. Is even
$$R_{ff}(\tau) = R_{ff}(-\tau)$$
   - Proof: change variable in definition
2. At infinite it goes to a constant
$$R_{ff}(\infty) = \overline{f(t)}^2 = const$$
   - Proof: two samples separated by ∞ are independent
3. Is maximum in 0
$$R_{ff}(0) \geq R_{ff}(\tau)$$
   - Proof: start from $\overline{(f(t) - f(t+\tau))^2} \geq 0$
   - Interpretation: different samples might vary differently, but a sample always varies identically with itself

Properties of the autocorrelation function

4. Value in 0 = the power of the random process

$$R_{ff}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{ff}(\omega) d\omega$$

   - Proof: Put $\tau = 0$ in inverse Fourier transform of Wiener-Khinchin theorem

5. Variance = difference between values at 0 and ∞

$$\sigma^2 = R_{ff}(0) - R_{ff}(\infty)$$

   - Proof: $R_{ff}(0) = \overline{f(t)^2}$, $R_{ff}(\infty) = \overline{f(t)}^2$

Autocorrelation of filtered random processes

- Consider a stationary random process applied as input to a LTI system
  - either continuous-time: input $x(t)$, system $H(s)$, output $y(t)$
  - or discrete-time: input $x[n]$, system $H(z)$, output $y[n]$
- How does the autocorrelation of y depend on that of the input x?
  - y is the convolution between x and the impulse response h

Computations

- For discrete-time processes

$$
\begin{aligned}
R_{yy}(\tau) &= \overline{y[n]y[n+\tau]} \\
&= \overline{\sum_{k_1=-\infty}^{\infty} h[k_1]x[n-k_1] \sum_{k_2=-\infty}^{\infty} h[k_2]x[n+\tau-k_2]} \\
&= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} h[k_1]h[k_2]\overline{x[n-k_1]x[n+\tau-k_2]} \\
&= \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} h[k_1]h[k_2]R_{xx}[\tau+k_1-k_2]
\end{aligned}
$$

- From Wiener-Hincin theorem:

$$
S_{ff}(\omega) = \sum_{\tau=-\infty}^{\infty} R_{ff}(\tau)e^{-j\omega\tau}
$$

Computations

- Therefore

$$S_{yy}(\omega) = \sum_{\tau=-\infty}^{\infty} \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} h[k_1]h[k_2]R_{xx}[\tau + k_1 - k_2]e^{-j\omega\tau}$$

- Change of variable: $\tau + k_1 - k_2 = u$
  - then $\tau = u - k_1 + k_2$

$$\begin{aligned}
S_{yy}(\omega) &= \sum_{u=-\infty}^{\infty} \sum_{k_1=-\infty}^{\infty} \sum_{k_2=-\infty}^{\infty} h[k_1]h[k_2]R_{xx}[u]e^{-j\omega(u-k_1+k_2)} \\
&= \sum_{u=-\infty}^{\infty} R_{xx}[u]e^{-j\omega u} \sum_{k_1=-\infty}^{\infty} h[k_1]e^{j\omega k_1} \sum_{k_2=-\infty}^{\infty} h[k_2]e^{-j\omega k_2} \\
&= S_{xx}(\omega) \cdot H^*(\omega) \cdot H(\omega) \\
&= S_{xx}(\omega) \cdot |H(\omega)|^2
\end{aligned}$$

Result

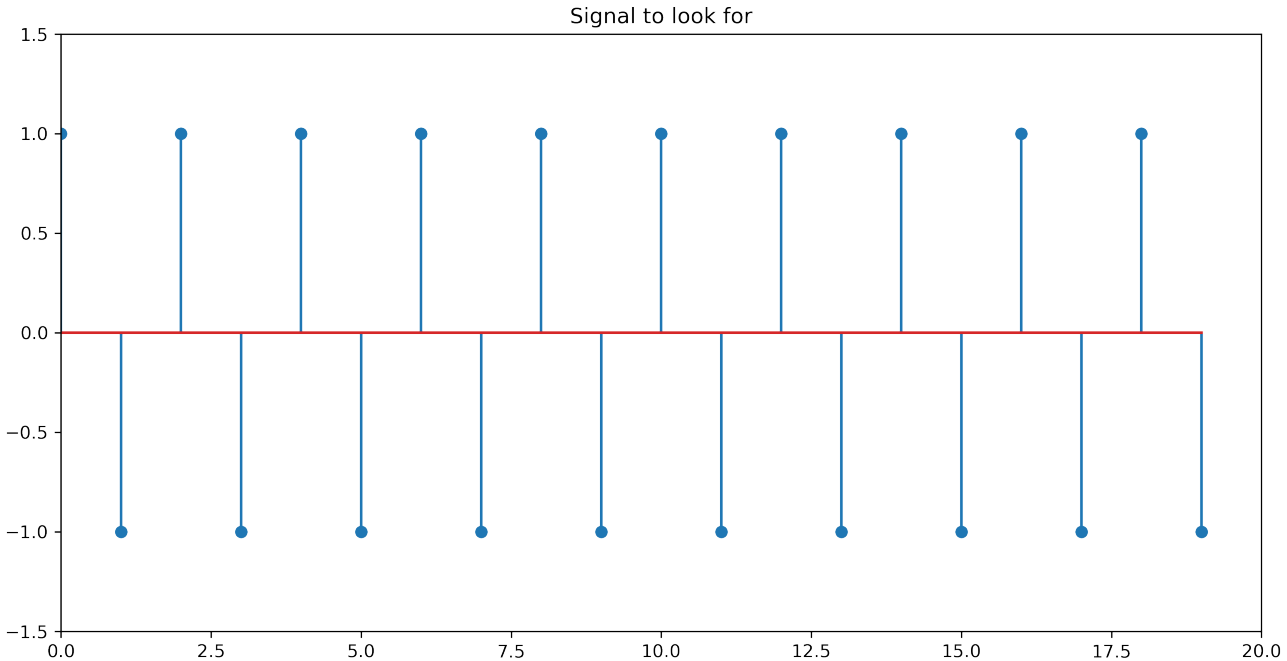
$$S_{yy}(\omega) = S_{xx}(\omega) \cdot |H(\omega)|^2$$

- The PSD of y = the PSD of x multiplied with the squared amplitude response of the filter
- Same relation is valid for continuous processes as well

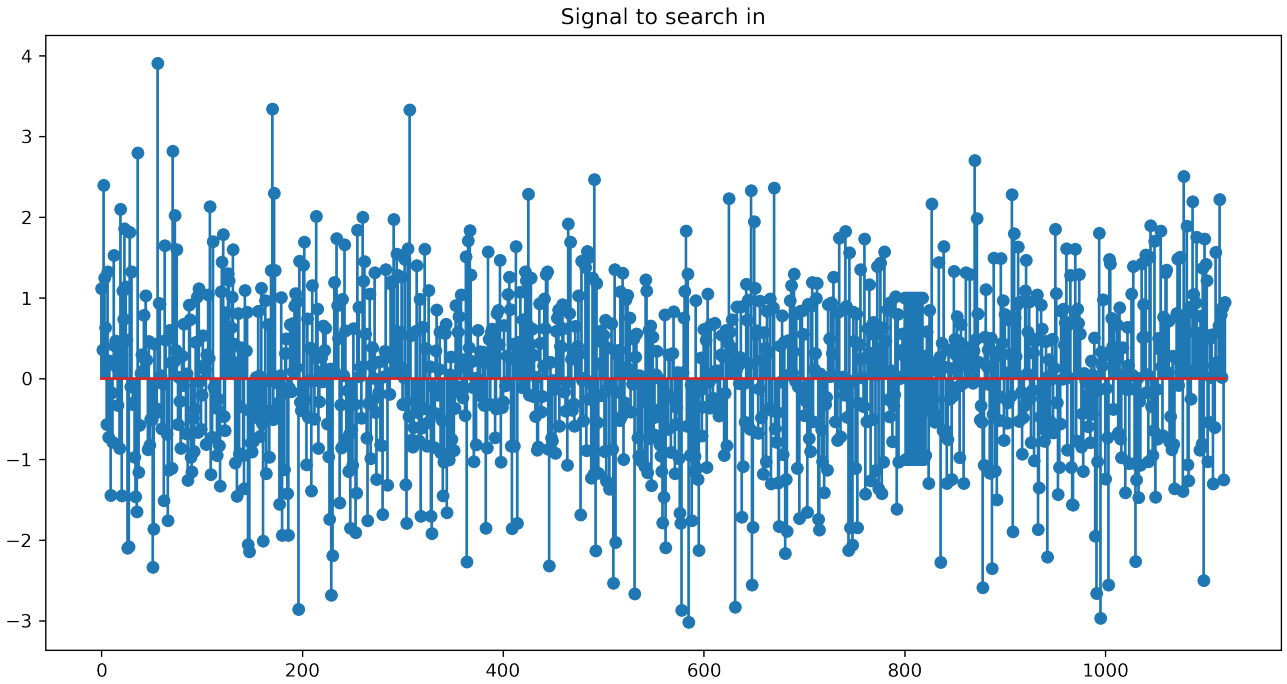
Applications of (auto)correlation

- Searching for a certain part in a large signal
- Correlation of two signals = measure of **similarity** of the two signals
  - The correlation function measures the similarity of a signal with all the shifted versions of the other
  - Example at blackboard
- Correlation can be used to locate data
  - The (auto)correlation function has large values when the two signals match
  - Large value when both positive and negative areas match,
  - Small values when they don't match

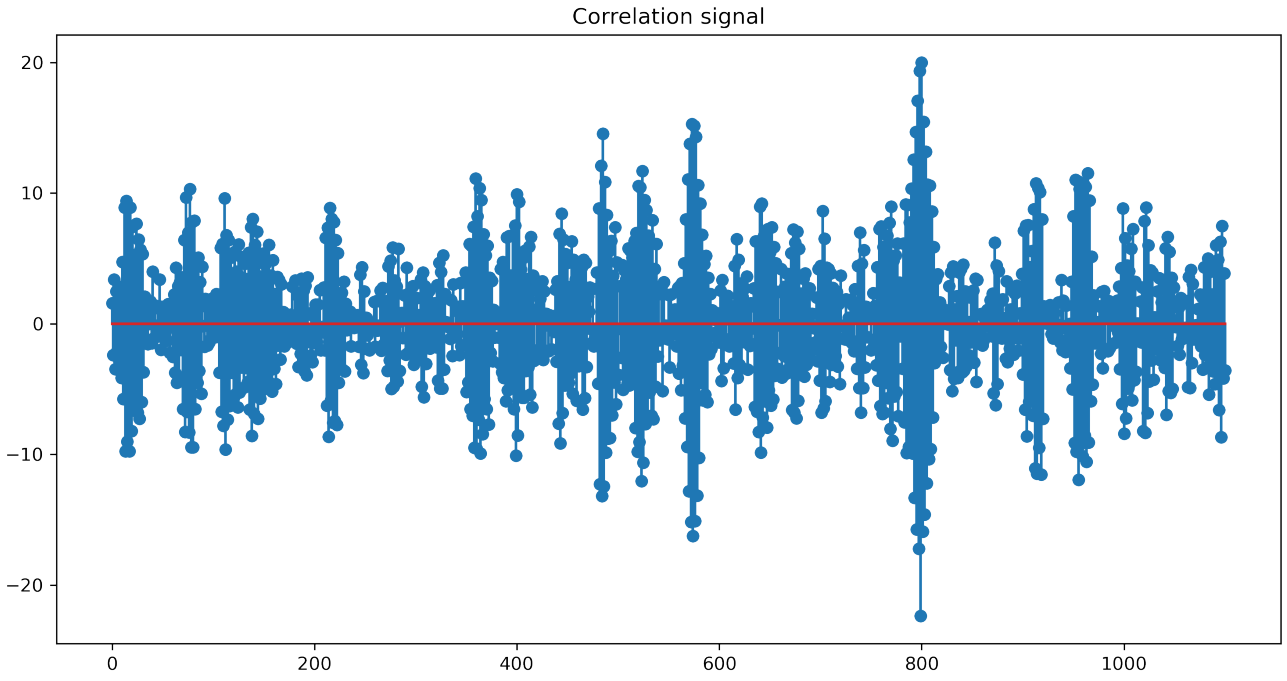
The signal to look for



The complete signal



Correlation result



System identification

- Determining the impulse response of an unknown LTI system
- Based on correlation between input and output of the system

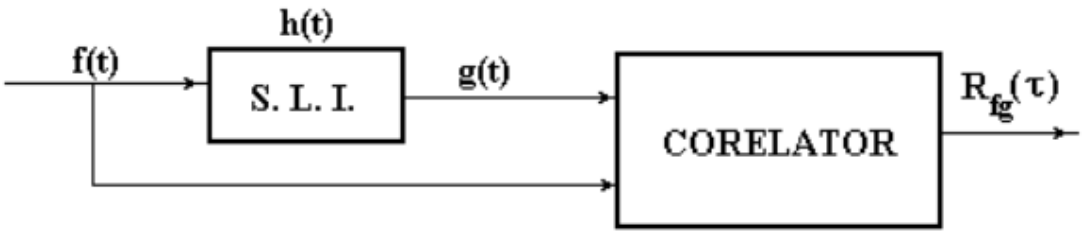


Figure 3: System identification setup

System identification

$$
\begin{aligned}
R_{fg}(\tau) =& \overline{f[n]g[n+\tau]} \\
=& \overline{f[n] \sum_{k=-\infty}^{\infty} h[k] f[n+\tau-k]} \\
=& \sum_{k=-\infty}^{\infty} h[k] \overline{f[n] f[n+\tau-k]} \\
=& \sum_{k=-\infty}^{\infty} h[k] R_{ff}[\tau-k] \\
=& h[\tau] \star R_{ff}[\tau]
\end{aligned}
$$

- If the input f is **white noise** with power A, $R_{ff}[n] = A \cdot \delta[n]$, and

$$R_{fg}(\tau) = h[\tau] \star R_{ff}[\tau] = A \cdot h[\tau] \star \delta[\tau] = A \cdot h[\tau]$$

- Then the correlation is proportional with the impulse response of the unknown system